# Four-Dimensional Context Hypergraphs with Separating Two-Valued States and a Chromatic Gap

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We construct context hypergraphs in four-dimensional real Hilbert space that admit a unital, separating family of two-valued states while their orthogonality graphs require five colors and have clique number four. The main example is a linear hypergraph with 65 rays and 29 contexts, obtained by completing a known 17-ray Peres subgraph. We give an injective integer realization with exactly the prescribed orthogonalities, prove the chromatic gap by a parity argument, and determine the fractional chromatic number and the full two-valued state space. A family of 48 states already provides a partition representation of the vertices and listed contexts. An elementary apex example and a hyperplane obstruction explain the role of the four-dimensional completion.

## I. INTRODUCTION AND OVERVIEW

Let $H = (V, E)$, with $E \neq \emptyset$, be a 4-uniform context hypergraph: each context $e \in E$ is a 4-element subset of $V$, and its orthogonality graph $\Gamma(H)$ has vertex set $V$ and edge set

$$E(\Gamma(H)) = \big\{\{u, v\} \subseteq V : u \neq v,\ \exists\, e \in E,\ \{u, v\} \subseteq e\big\}. \tag{1}$$

Writing $K_t$ for the complete graph on $t$ vertices, every context $e \in E$ forms a 4-clique ($K_4$) in $\Gamma(H)$, so the clique number satisfies $\omega(\Gamma(H)) \geq 4$, and equality $\omega(\Gamma(H)) = 4$ holds whenever $\Gamma(H)$ is $K_5$-free. A *strong $k$-coloring* of $H$ is a proper $k$-coloring $c : V \to \{0, 1, \ldots, k-1\}$ of $\Gamma(H)$, so that every context $e \in E$ receives four distinct colors; the minimum such $k$ is the chromatic number $\chi(H) = \chi(\Gamma(H)) \geq \omega(\Gamma(H))$.

In four-dimensional Hilbert space, a 4-uniform hypergraph $H = (V, E)$ is *edge-faithfully coordinatizable in* $\mathbb{R}^4$ if there is an injective ray assignment $i \mapsto [v_i] \in \mathbb{P}(\mathbb{R}^4)$, the projective space of rays in real Euclidean four-space ($v_i \in \mathbb{R}^4 \setminus \{0\}$) such that

$$v_i \cdot v_j = 0 \iff \{i, j\} \in E(\Gamma(H)) \qquad (i \neq j \in V). \tag{2}$$

In particular, every context $e \in E$ is mapped to an orthogonal basis of $\mathbb{R}^4$, no two vertices share a ray, and no extra orthogonalities occur outside $E(\Gamma(H))$.

Independently of quantum coordinatization, we call $H$ *set-representable on its vertices and listed contexts* if its collection $\mathcal{S}(H)$ of *two-valued states*—maps $s : V \to \{0, 1\}$ satisfying $\sum_{i \in e} s(i) = 1$ for every $e \in E$—is both:

1. *unital*: for every $i \in V$, there exists $s \in \mathcal{S}(H)$ with $s(i) = 1$; and

2. *separating*: for every pair of distinct vertices $i \neq j \in V$, there exists $s \in \mathcal{S}(H)$ with $s(i) \neq s(j)$.

For any unital and separating family $\Omega \subseteq \mathcal{S}(H)$, set

$$A_i = \{s \in \Omega : s(i) = 1\} \qquad (i \in V). \tag{3}$$

The sets $A_i$ are nonempty and pairwise distinct, and $\{A_i : i \in e\}$ is a partition of $\Omega$ for each $e \in E$. This is the precise partition representation meant here [1, 2]; it does not require injectivity on all compound Hilbert-space events. The support of a state is $\mathrm{supp}(s) = \{i \in V : s(i) = 1\}$.

The *independence number* $\alpha(G)$ is the largest size of a vertex set containing no adjacent pair. The *fractional chromatic number* $\chi_f(G)$ is the minimum total nonnegative weight assigned to independent sets so that the sets containing each vertex have total weight at least one.

Because the color classes of any strong 4-coloring of a 4-uniform hypergraph define four two-valued states that partition $V$, chromatic completeness ($\chi(H) = 4$) implies unitality, whereas Kochen–Specker (KS) configurations such as the 24-ray Peres set in $\mathbb{R}^4$ [3] satisfy $\chi > 4$ because $\mathcal{S}(H) = \emptyset$. The subscript YO abbreviates Yu–Oh. Conversely, the existence of unital and separating states need not supply a strong 4-coloring: such a coloring requires four states whose supports are pairwise disjoint and cover $V$.

The purpose of this note is to exhibit explicit 4-uniform context hypergraphs $H$ with $\omega(\Gamma(H)) = 4$ that are simultaneously set-representable (unital and separating two-valued states), edge-faithfully coordinatizable in $\mathbb{R}^4$, and chromatically incomplete ($\chi(H) = 5 > 4 = \omega(\Gamma(H))$). Table I summarizes the configurations analyzed below.

## II. NON-LINEAR WARM-UP $H_{\mathrm{YO}}^{(4)}$ AND THE HYPERPLANE OBSTRUCTION

Before constructing a linear 4-uniform example, we record an elementary 26-ray, 16-context non-linear example obtained by lifting the 3-uniform completed Yu–Oh hypergraph $H_{\mathrm{YO}}^{+} = (V_{\mathrm{YO}}^{+}, E_{\mathrm{YO}}^{+})$ [4–6] from $\mathbb{R}^3$ to $\mathbb{R}^4$. The obstruction below rules out retaining linearity when two intersecting complete orthogonal triads in one hyperplane are lifted to distinct four-element contexts.

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TABLE I. Comparison of 4-uniform context hypergraphs in $\mathbb{R}^4$ with clique number $\omega(\Gamma(H)) = 4$ and chromatic number $\chi(H) = 5 > 4$, showing the number of pairwise non-collinear vectors $|V|$, the number of 4-element contexts $|E|$, and the number of orthogonality edges $|E(\Gamma(H))|$.

| Hypergraph | $\lvert V\rvert$ | $\lvert E\rvert$ | $\lvert E(\Gamma(H))\rvert$ | Linear | $\omega(\Gamma(H))$ | $\chi(H)$ | Edge-faithful | $\lvert\mathcal{S}(H)\rvert$ | Set-representable |
|---|---|---|---|---|---|---|---|---|---|
| Peres 24-ray set [3] | 24 | 24 | 108 | No ($\lvert e\cap e'\rvert \le 2$) | 4 | 5 | Yes ($\mathbb{R}^4$) | 0 | No (KS) |
| Apex suspension $H^{(4)}_{\mathrm{YO}}$ (Prop. 1) | 26 | 16 | 73 | No ($\lvert e\cap e'\rvert \le 2$) | 4 | 5 | Yes ($\mathbb{R}^4$) | 25 | Yes |
| Canonical linear $H_{65}$ (Thm. 4) | 65 | 29 | 174 | Yes ($\lvert e\cap e'\rvert \le 1$) | 4 | 5 | Yes ($\mathbb{R}^4$) | 393,216 | Yes (48 canonical) |

**Proposition 1** (Non-linear apex suspension $H^{(4)}_{\mathrm{YO}}$)**.** *Let $H^{+}_{\mathrm{YO}} = (V^{+}_{\mathrm{YO}}, E^{+}_{\mathrm{YO}})$ be the 25-vertex, 16-context completed Yu–Oh hypergraph in $\mathbb{R}^3$, with vectors $r_1, \dots, r_{25} \in \mathbb{R}^3$ and contexts $E^{+}_{\mathrm{YO}} = \{e_1, \dots, e_{16}\}$. Define $H^{(4)}_{\mathrm{YO}} = (V_{26}, E^{(4)}_{16})$ on $V_{26} = \{1, \dots, 26\}$ by appending an apex vertex 26 to every context:*

$$E^{(4)}_{16} = \big\{e \cup \{26\} : e \in E^{+}_{\mathrm{YO}}\big\}, \tag{4}$$

*and assign the vectors $v_i = (r_i, 0)^T \in \mathbb{R}^4$ for $1 \le i \le 25$ and $v_{26} = (0, 0, 0, 1)^T \in \mathbb{R}^4$ (enumerated explicitly in Appendix A 2 and depicted as a 4-uniform hypergraph diagram in Fig. 1). Then:*

1. *$H^{(4)}_{\mathrm{YO}}$ is 4-uniform with clique number $\omega(\Gamma(H^{(4)}_{\mathrm{YO}})) = 4$ and chromatic number $\chi(H^{(4)}_{\mathrm{YO}}) = \chi(H^{+}_{\mathrm{YO}}) + 1 = 5 > 4$.*
2. *The vectors $v_1, \dots, v_{26} \in \mathbb{R}^4$ form an injective, edge-faithful orthogonal representation of $H^{(4)}_{\mathrm{YO}}$ in $\mathbb{R}^4$.*
3. *$H^{(4)}_{\mathrm{YO}}$ admits exactly 25 two-valued states (enumerated in Appendix B 1), which form a unital and separating collection on $V_{26}$.*

*Proof.* (1)–(2) Since $v_{26} = (0, 0, 0, 1)^T$ is orthogonal to $\mathbb{R}^3 \times \{0\}$ and $r_1, \dots, r_{25}$ is an injective, edge-faithful representation of $H^{+}_{\mathrm{YO}}$ in $\mathbb{R}^3$, the vectors $v_1, \dots, v_{26}$ are pairwise non-proportional and satisfy $v_i \cdot v_j = 0$ if and only if $\{i, j\} \in E(\Gamma(H^{(4)}_{\mathrm{YO}}))$. Thus $\Gamma(H^{(4)}_{\mathrm{YO}})$ is the graph join $\Gamma(H^{+}_{\mathrm{YO}}) * K_1$, which immediately gives $\omega(\Gamma(H^{(4)}_{\mathrm{YO}})) = \omega(\Gamma(H^{+}_{\mathrm{YO}})) + 1 = 3 + 1 = 4$ and $\chi(H^{(4)}_{\mathrm{YO}}) = \chi(H^{+}_{\mathrm{YO}}) + 1 = 4 + 1 = 5$.

(3) In any two-valued state $s : V_{26} \to \{0, 1\}$ on $H^{(4)}_{\mathrm{YO}}$, either $s(26) = 1$, which forces $s(1) = \cdots = s(25) = 0$ (giving one apex state $s_{\mathrm{apex}}$), or $s(26) = 0$, in which case $\sum_{i \in e} s(i) = 1$ for all $e \in E^{+}_{\mathrm{YO}}$ so that $s|_{V^{+}_{\mathrm{YO}}}$ is one of the 24 two-valued states of $H^{+}_{\mathrm{YO}}$. Since the 24 states of $H^{+}_{\mathrm{YO}}$ are unital and separating on $\{1, \dots, 25\}$ and all vanish at 26, while $s_{\mathrm{apex}}(26) = 1$ and $s_{\mathrm{apex}}(i) = 0$ for $i \le 25$, the 25 states of $H^{(4)}_{\mathrm{YO}}$ are unital and separating on all 26 vertices. □

Although $H^{(4)}_{\mathrm{YO}}$ satisfies all three requirements of set-representability, edge-faithfulness in $\mathbb{R}^4$, and $\chi = 5 > 4 = \omega$, it is *non-linear*: the apex vertex 26 lies in all 16 contexts, so $|(e \cup \{26\}) \cap (e' \cup \{26\})| = 2$ whenever $e, e' \in E^{+}_{\mathrm{YO}}$ share a vertex in $\mathbb{R}^3$ (giving Berge girth 2). A Berge cycle of length $r$ consists of $r$ distinct contexts and $r$ distinct vertices, with successive contexts meeting at the corresponding vertices cyclically; the Berge girth is the shortest such length. The following proposition shows that this non-linearity is unavoidable whenever two contexts have intersecting three-dimensional restrictions inside a common hyperplane.

**Proposition 2** (Hyperplane obstruction to linearity in dimension four)**.** *Let $H = (V, E)$ be a 4-uniform context hypergraph with an injective orthogonal representation $i \mapsto [v_i] \in \mathbb{P}(\mathbb{F}^4)$ ($\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, with $\mathbb{C}$ the complex numbers). Orthogonality over $\mathbb{C}$ is defined by the Hermitian product $\langle x, y\rangle = \sum_{j=1}^{4} \overline{x_j} y_j$. For a subspace $W$, write $e_W = \{i \in e : v_i \in W\}$. If two distinct contexts $e, e' \in E$ each contain at least three vertices whose rays lie in a common three-dimensional hyperplane $W = z^{\perp} \subset \mathbb{F}^4$ ($z \in \mathbb{F}^4 \setminus \{0\}$), then both $e$ and $e'$ must contain the unique ray $[z]$ orthogonal to $W$. Consequently, if $|e_W \cap e'_W| \ge 1$, then $|e \cap e'| \ge 2$, violating linearity.*

*Proof.* Let $e = \{u_1, u_2, u_3, u_4\}$ with $v_{u_1}, v_{u_2}, v_{u_3} \in W$. Because $v_{u_1}, v_{u_2}, v_{u_3}$ are mutually orthogonal nonzero vectors in the 3-dimensional subspace $W$, they form an orthogonal basis of $W$. Since $v_{u_4} \perp v_{u_1}, v_{u_2}, v_{u_3}$, we have $v_{u_4} \in W^{\perp} = \mathrm{span}(z)$, so $[v_{u_4}] = [z]$. The same argument applied to $e'$ shows that its fourth ray must also equal $[z]$; by injectivity, the same vertex $z_0 \in V$ with $[v_{z_0}] = [z]$ belongs to both $e$ and $e'$. If $e$ and $e'$ also share any vertex inside $W$, then $|e \cap e'| \ge 2$. □

## III. CONSTRUCTION OF THE CANONICAL LINEAR HYPERGRAPH $H_{65}$ IN $\mathbb{R}^4$

By Proposition 2, two intersecting complete orthogonal triads in one hyperplane cannot be extended to distinct contexts of a linear 4-uniform hypergraph with an injective orthogonal representation in $\mathbb{R}^4$. The assertion concerns triads, not arbitrary smaller restrictions of contexts. Instead, we start from a four-dimensional core of 17 rays in $\{-1, 0, 1\}^4$ and complete its 24 uncovered orthogonal pairs into 4-contexts via the following general lemma. All completion vectors are integer vectors, with $\mathbb{Z}$ denoting the integers.

### A. The two-plane completion lemma in $\mathbb{R}^4$

In $\mathbb{R}^3$, two orthogonal rays $\{v_u, v_v\}$ uniquely determine their third orthogonal basis ray $[v_u \times v_v]$, which may inadvertently be orthogonal or parallel to other rays of the configuration. In $\mathbb{R}^4$, by contrast, the orthogonal complement $\{v_u, v_v\}^\perp$ of an orthogonal pair is a *two-dimensional plane*, providing an infinite one-parameter family of orthogonal completion pairs $\{w_1, w_2\}$.

**Lemma 3** (Simultaneous plane-completion faithfulness and state extension in $\mathbb{R}^4$)**.** *Let $V_{\text{core}}$ be a finite vertex set carrying pairwise non-collinear vectors $v_i \in \mathbb{Q}^4 \setminus \{0\}$, where $\mathbb{Q}$ denotes the rational numbers, let $E_{\text{base}}$ be a collection of 4-element orthogonal bases of $\mathbb{R}^4$ drawn from $V_{\text{core}}$ with $|C \cap C'| \le 1$ for all $C \neq C' \in E_{\text{base}}$, and let*

$$E_{\text{bridge}} = \big\{\{u,v\} \subset V_{\text{core}} : v_u \cdot v_v = 0\big\} \setminus \bigcup_{C \in E_{\text{base}}} \binom{C}{2} \quad (5)$$

*be the set of orthogonal pairs in $V_{\text{core}}$ not already covered by $E_{\text{base}}$, with $E_{\text{bridge}} = \{e_1, \dots, e_M\}$ ($e_m = \{u_m, v_m\}$). For each $m \in \{1, \dots, M\}$, set $L_m = \operatorname{span}(v_{u_m}, v_{v_m})$ and $P_m = L_m^\perp$.*

1. Span conditions for simultaneous edge-faithful completion: *The pairs $e_1, \dots, e_M \in E_{\text{bridge}}$ can be simultaneously completed to 4-element orthogonal bases $E_m = \{u_m, v_m, w_{m,1}, w_{m,2}\}$ by fresh integer vectors $w_{m,1}, w_{m,2} \in P_m \cap \mathbb{Z}^4$ having* no *extra orthogonalities or collinearities with any other vectors in the completed hypergraph $H = (V_{\text{core}} \cup \bigcup_{m=1}^M \{w_{m,1}, w_{m,2}\},\ E_{\text{base}} \cup \{E_1, \dots, E_M\})$ if and only if both*

$$v_z \notin L_m \quad \text{for every } z \in V_{\text{core}} \setminus \{u_m, v_m\} \quad (1 \le m \le M) \quad (6)$$

*and*

$$P_\ell \neq L_m \ (\text{i.e., } L_\ell \not\perp L_m) \quad (1 \le \ell \neq m \le M). \quad (7)$$

2. Linearity, clique number, and chromatic preservation: *The completed 4-uniform hypergraph $H$ is linear ($|e \cap e'| \le 1$ for all $e \neq e' \in E(H)$). If $E(H) \neq \emptyset$ and the core orthogonality graph $G_{\text{core}}$ is $K_5$-free, then $\omega(\Gamma(H)) = 4$, and for every $k \ge 4$, $H$ is strongly $k$-colorable if and only if $G_{\text{core}}$ is properly $k$-colorable.*

3. Canonical state extension: *Let $S_1, \dots, S_N \subset V_{\text{core}}$ be all independent sets in $G_{\text{core}}$ satisfying $|S_\alpha \cap C| = 1$ for every $C \in E_{\text{base}}$. For each $\alpha \in \{1, \dots, N\}$ and $\varepsilon \in \{0,1\}$, define $s_{\alpha,\varepsilon} : V(H) \to \{0,1\}$ by $s_{\alpha,\varepsilon}(i) = \mathbf{1}_{i \in S_\alpha}$ for $i \in V_{\text{core}}$ and, for each $m \in \{1, \dots, M\}$,*

$$\big(s_{\alpha,\varepsilon}(w_{m,1}),\ s_{\alpha,\varepsilon}(w_{m,2})\big) = \begin{cases} (0,0), & |S_\alpha \cap \{u_m, v_m\}| = 1, \\ (1-\varepsilon,\ \varepsilon), & S_\alpha \cap \{u_m, v_m\} = \emptyset. \end{cases} \quad (8)$$

*Then every $s_{\alpha,\varepsilon}$ is a valid two-valued state on $H$; here $\mathbf{1}_{i \in S_\alpha}$ is the indicator of membership in $S_\alpha$. The two indices $\varepsilon = 0, 1$ can give the same state when $S_\alpha$ hits every bridge edge. If $S_1, \dots, S_N$ are unital and separating on $V_{\text{core}}$, and the $M$ edge zero-profiles*

$$Z(e_m) = \big\{\alpha \in \{1, \dots, N\} : u_m \notin S_\alpha \text{ and } v_m \notin S_\alpha\big\} \quad (9)$$

*are nonempty and pairwise distinct for $m = 1, \dots, M$, then the indexed canonical family $\{s_{\alpha,\varepsilon}\}$, containing at most $2N$ distinct states, is unital and separating on all $|V_{\text{core}}| + 2M$ vertices of $H$.*

*Proof.* (1) For necessity, if $v_z \in L_m$ for some $z \in V_{\text{core}} \setminus \{u_m, v_m\}$, then $P_m = L_m^\perp \subset v_z^\perp$, so every completion vector in $P_m$ is orthogonal to $v_z$; similarly, if $P_\ell = L_m$ for $\ell \neq m$, then $P_\ell \perp P_m$, forcing $w_{\ell,a} \perp w_{m,b}$ for all $a, b \in \{1,2\}$.

For sufficiency, assume Eqs. (6)–(7). Because $v_{u_m}, v_{v_m} \in \mathbb{Q}^4$, each $P_m$ is a rational 2-plane and therefore admits an integer orthogonal basis $\{\eta_A, \eta_B\} \subset P_m \cap \mathbb{Z}^4$ (obtained by rational Gram–Schmidt and clearing denominators) with $n_A = \|\eta_A\|^2$ and $n_B = \|\eta_B\|^2$. We choose $\{w_{m,1}, w_{m,2}\} \subset P_m \cap \mathbb{Z}^4$ inductively for $m = 1, \dots, M$ so as to maintain the invariant that, for every $r \ge m$, no previously placed vector $z \notin \{v_{u_r}, v_{v_r}\}$ lies in $L_r$ (which holds at $m = 1$ by Eq. (6)). At step $m$, the invariant for $r = m$ ensures that every previously placed vector $z \notin \{v_{u_m}, v_{v_m}\}$ satisfies $z \notin L_m = P_m^\perp$, so $z^\perp \cap P_m$ and $\mathbb{R}z \cap P_m$ are at most 1-dimensional lines in $P_m$. Moreover, for every future $r > m$, Eq. (7) gives $P_m \neq L_r$, so $P_m \cap L_r$ is also at most a 1-dimensional line in $P_m$. The integer parameterization

$$\begin{aligned} w_{m,1}(a,b) &= a\,\eta_A + b\,\eta_B, \\ w_{m,2}(a,b) &= -n_B b\,\eta_A + n_A a\,\eta_B \end{aligned} \quad (10)$$

produces infinitely many projectively distinct orthogonal pairs in $P_m \cap \mathbb{Z}^4$ as $(a,b)$ ranges over $\mathbb{Z}^2 \setminus \{(0,0)\}$, and can avoid every forbidden line for both vectors: the inverse image of each such line under either invertible linear map in Eq. (10) is a line in the $(a,b)$-parameter plane, and finitely many lines cannot cover all integer slopes. This avoids the finite union of forbidden lines in $P_m$, preventing any extra orthogonalities or collinearities with previously placed vectors while preserving $w_{m,1}, w_{m,2} \notin L_r$ for all $r > m$.

(2) Each $E_m$ shares at most one vertex with any $C \in E_{\text{base}}$ (since $\binom{C}{2} \cap E_{\text{bridge}} = \emptyset$) and at most one vertex with any $E_{m'}$ ($m' \neq m$), while $w_{m,1}, w_{m,2}$ have degree 1 in $H$. In any proper $k$-coloring ($k \ge 4$) of $G_{\text{core}}$, $c(u_m) \neq c(v_m)$, leaving $k - 2 \ge 2$ colors in $\{0, \dots, k-1\} \setminus \{c(u_m), c(v_m)\}$ to color $w_{m,1}$ and $w_{m,2}$.

(3) Since $S_\alpha$ is independent in $G_{\text{core}}$, $|S_\alpha \cap \{u_m, v_m\}| \in \{0,1\}$, so Eq. (8) gives $s_{\alpha,\varepsilon}(u_m) + s_{\alpha,\varepsilon}(v_m) + s_{\alpha,\varepsilon}(w_{m,1}) + s_{\alpha,\varepsilon}(w_{m,2}) = 1$ on every $E_m$. Across the $2N$ indexed states $(s_{1,0}, s_{1,1}, \dots, s_{N,0}, s_{N,1})$, the profile of $w_{m,1}$ has $(1,0)$ in block $\alpha$ for $\alpha \in Z(e_m)$ and $(0,0)$ otherwise; the

profile of $w_{m,2}$ has $(0,1)$ in block $\alpha$ for $\alpha \in Z(e_m)$ and $(0,0)$ otherwise; and the profile of $i \in V_{\text{core}}$ has $(1,1)$ for $i \in S_\alpha$ and $(0,0)$ otherwise. Since $Z(e_m) \neq \emptyset$ and $Z(e_m) \neq Z(e_{m'})$ for $m \neq m'$, all $|V_{\text{core}}|+2M$ state profiles are nonzero and pairwise distinct. □

### B. Explicit definition of $H_{65}$ and provenance of the 17-ray core

We define the 17-vertex core $V_{\text{core}} = \{1, \ldots, 17\}$ in $\mathbb{R}^4$ using four structural groups of vectors in $\{-1, 0, 1\}^4$:

1. *Standard coordinate basis* $C_1 = \{1, 2, 3, 4\}$*:*

$$v_1 = (1,0,0,0), \qquad v_2 = (0,1,0,0),$$
$$v_3 = (0,0,1,0), \qquad v_4 = (0,0,0,1).$$

2. *Even-parity Hadamard basis* $C_2 = \{5, 6, 7, 8\}$ (the ray representatives with first coordinate $+1$ and an even number of minus signs):

$$v_5 = (1,1,1,1), \qquad v_6 = (1,1,-1,-1),$$
$$v_7 = (1,-1,1,-1), \qquad v_8 = (1,-1,-1,1).$$

3. *Odd-parity Hadamard hub triad* $\{9, 10, 11\}$ (three mutually orthogonal $\pm 1$ vectors with an odd number of minus signs):

$$v_9 = (1,1,-1,1), \qquad v_{10} = (1,-1,1,1),$$
$$v_{11} = (1,-1,-1,-1).$$

4. *Three binary bridge pairs* $B_1 = \{12, 13\}$, $B_2 = \{14, 15\}$, $B_3 = \{16, 17\}$ completing the three pairs of $\{9, 10, 11\}$ into 4-element orthogonal bases $C_3 = \{9, 10, 12, 13\}$, $C_4 = \{9, 11, 14, 15\}$, and $C_5 = \{10, 11, 16, 17\}$:

$$v_{12} = (1,0,0,-1), \qquad v_{13} = (0,1,1,0),$$
$$v_{14} = (1,0,1,0), \qquad v_{15} = (0,1,0,-1),$$
$$v_{16} = (1,1,0,0), \qquad v_{17} = (0,0,1,-1).$$

As a 17-ray orthogonality graph in $\mathbb{R}^4$, $G_{\text{core}}$ is orthogonally equivalent to the 17-ray subgraph of the 24-ray Peres configuration introduced by Haynes *et al.* [7, Sec. 1.2, pp. 8–9] as an example with orthogonal dimension 4 and chromatic number 5. Explicitly, the orthogonal matrix

$$T = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & -1 \\ 1 & 1 & -1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & -1 & -1 & -1 \end{pmatrix}, \qquad T^T T = I_4, \tag{11}$$

where $I_4$ is the $4 \times 4$ identity matrix. The matrix $T$ maps the hub triad $\{v_9, v_{10}, v_{11}\}$ to the coordinate rays $\{e_2, e_3, e_4\}$, the bases $C_1 \cup C_2 = \{v_1, \ldots, v_8\}$ to the eight rays $(1, \pm 1, \pm 1, \pm 1)^T$, and the six bridge rays $\{v_{12}, \ldots, v_{17}\}$ to the six binary rays $e_1 \pm e_j$ $(j = 2, 3, 4)$ of Ref. [7]. Choosing the coordinate-Hadamard frame above makes the bipartite parity structure between $C_1$, $C_2$, and $B_1 \cup B_2 \cup B_3$ transparent, allowing us to complete $G_{\text{core}}$ into the linear 4-uniform context hypergraph $H_{65}$ and analyze its two-valued state space by hand.

| $i$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $x_{i,1}$ | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 |
| $x_{i,2}$ | 0 | 1 | 0 | 0 | 1 | 1 | −1 | −1 | 1 | −1 | −1 | 0 | 1 | 0 | 1 | 1 | 0 |
| $x_{i,3}$ | 0 | 0 | 1 | 0 | 1 | −1 | 1 | −1 | −1 | 1 | −1 | 0 | 1 | 1 | 0 | 0 | 1 |
| $x_{i,4}$ | 0 | 0 | 0 | 1 | 1 | −1 | −1 | 1 | 1 | 1 | −1 | −1 | 0 | 0 | −1 | 0 | −1 |
| $c(i)$ | 0 | 1 | 2 | 3 | 0 | 1 | 2 | 4 | 0 | 1 | 3 | 3 | 4 | 2 | 1 | 0 | 2 |

| $i$ | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $x_{i,1}$ | 1 | 3 | 1 | 3 | 0 | 0 | 2 | 3 | 1 | 3 | 1 | 3 | 0 | 0 | 2 | 3 | 1 | 3 | 1 | 3 | 0 | 0 | 2 | 3 |
| $x_{i,2}$ | 0 | 0 | 3 | −2 | 1 | 3 | −3 | 1 | 0 | 0 | −3 | 2 | 1 | 3 | 3 | −1 | −1 | −3 | −1 | −3 | 2 | 3 | 0 | 0 |
| $x_{i,3}$ | −3 | 2 | 0 | 0 | −1 | −3 | 3 | −1 | −1 | −3 | −1 | −3 | 3 | −2 | 0 | 0 | 0 | 0 | 3 | −2 | 3 | −1 | −3 | 1 |
| $x_{i,4}$ | 1 | 3 | 1 | 3 | 3 | −2 | 0 | 0 | −3 | 2 | 0 | 0 | 1 | 3 | 3 | −1 | 3 | −2 | 0 | 0 | 3 | −1 | −3 | 1 |
| $c(i)$ | 0 | 2 | 0 | 1 | 1 | 2 | 0 | 1 | 0 | 3 | 0 | 1 | 2 | 3 | 0 | 3 | 1 | 3 | 1 | 2 | 1 | 3 | 0 | 3 |

| $i$ | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $x_{i,1}$ | 1 | 2 | 1 | 2 | 1 | 5 | 3 | 3 | 1 | 2 | 1 | 2 | 3 | 3 | 1 | 5 | 1 | 2 | 1 | 2 | 1 | 5 | 3 | 3 |
| $x_{i,2}$ | −5 | −1 | −3 | 3 | −2 | −1 | −1 | 2 | 3 | −3 | 5 | 1 | −2 | 1 | 2 | 1 | −1 | −2 | −1 | −2 | 3 | −3 | −5 | 1 |
| $x_{i,3}$ | 3 | −3 | 5 | 1 | 2 | 1 | 1 | −2 | −1 | −2 | −1 | −2 | 1 | −5 | 3 | −3 | −5 | −1 | −3 | 3 | −2 | −1 | −1 | 2 |
| $x_{i,4}$ | 1 | 2 | 1 | 2 | −3 | 3 | 5 | −1 | 5 | 1 | 3 | −3 | −2 | 1 | 2 | 1 | −3 | 3 | −5 | −1 | −2 | −1 | −1 | 2 |
| $c(i)$ | 1 | 2 | 0 | 1 | 0 | 2 | 0 | 1 | 0 | 3 | 0 | 1 | 2 | 3 | 0 | 3 | 1 | 3 | 1 | 2 | 1 | 3 | 0 | 3 |

(12)

Here the weight of a vector is its number of nonzero coordinates, and $N(i) = \{j \in V_{\text{core}} \setminus \{i\} : v_i \cdot v_j = 0\}$ denotes its neighborhood in $G_{\text{core}}$. Because every vector in $C_1$ has weight 1 and every vector in $C_2 \cup \{9, 10, 11\}$ has weight 4 with odd/even parity, there are no orthogonalities among $\{1, \ldots, 11\}$ outside $C_1$, $C_2$, and the triad $\{9, 10, 11\}$. Each of the six binary bridge vectors $b \in \{12, \ldots, 17\}$, having weight 2, is orthogonal to exactly two coordinate rays in $C_1$ and two even-parity Hadamard rays in $C_2$, creating $6 \times 2 + 6 \times 2 = 24$ bridge edges $E_{\text{bridge}} = \{e_1, \ldots, e_{24}\}$. Completing each $e_m = \{u_m, v_m\}$ via Lemma 3 with fresh integer rays $\{2m + 16,\ 2m + 17\} \subset \{18, \ldots, 65\}$ yields the 29 contexts of $H_{65} = (V_{65}, E_{29})$:

$$
\begin{aligned}
E_{29} = \big\{ & \{1,2,3,4\},\ \{5,6,7,8\},\ \{9,10,12,13\},\\
& \{9,11,14,15\},\ \{10,11,16,17\},\ \{2,12,18,19\},\\
& \{3,12,20,21\},\ \{1,13,22,23\},\ \{4,13,24,25\},\\
& \{2,14,26,27\},\ \{4,14,28,29\},\ \{1,15,30,31\},\\
& \{3,15,32,33\},\ \{3,16,34,35\},\ \{4,16,36,37\},\\
& \{1,17,38,39\},\ \{2,17,40,41\},\ \{5,12,42,43\},\\
& \{8,12,44,45\},\ \{6,13,46,47\},\ \{7,13,48,49\},\\
& \{6,14,50,51\},\ \{8,14,52,53\},\ \{5,15,54,55\},\\
& \{7,15,56,57\},\ \{7,16,58,59\},\ \{8,16,60,61\},\\
& \{5,17,62,63\},\ \{6,17,64,65\}\big\}.
\end{aligned}
\tag{13}
$$

The complete coordinate table $v_i = (x_{i,1}, x_{i,2}, x_{i,3}, x_{i,4})^T \in \mathbb{R}^4$ and a strong 5-coloring $c(i) \in \{0, 1, 2, 3, 4\}$ for all 65 vertices are given in Eq. (12) (and enumerated vertex-by-vertex and context-by-context in Appendix A 1). Figure 1 illustrates the 26-vertex, 16-context non-linear warm-up apex suspension $H_{\text{YO}}^{(4)}$ of Proposition 1 (with the common apex vertex 26 at the center surrounded by 16 concentric colored disks), while Fig. 2 presents the full 65-vertex, 29-context four-uniform hypergraph diagram of the canonical linear hypergraph $H_{65}$.

Both drawings represent 4-uniform context hypergraphs. They differ from Greechie diagrams in the strict sense of Ref. [1, Definition 2.4]: the warm-up has contexts sharing two vertices, whereas $H_{65}$ contains the three-context Berge cycle $C_3, C_4, C_5$, with successive intersection vertices $9, 11, 10$. Strict Greechie diagrams permit neither feature. The smooth-curve drawing convention is used only to display the listed incidences.

## IV. MAIN THEOREM AND PROOFS

**Theorem 4** (Properties of the canonical linear 4-uniform hypergraph $H_{65}$)**.** *The 65-vertex, 29-context 4-uniform hypergraph $H_{65} = (V_{65}, E_{29})$ defined by Eqs. (12)–(13) satisfies:*

1. Linearity and clique number: *$H_{65}$ is linear ($|e \cap e'| \le 1$ for all distinct $e, e' \in E_{29}$), and the 4-cliques of $\Gamma(H_{65})$ are precisely the 29 contexts of $E_{29}$; in particular, $\omega(\Gamma(H_{65})) = 4$.*

2. Edge-faithful realization in $\mathbb{R}^4$: *The 65 integer vectors $v_1, \ldots, v_{65} \in \mathbb{R}^4$ in Eq. (12) are pairwise non-collinear and satisfy $v_i \cdot v_j = 0$ if and only if $\{i, j\} \in E(\Gamma(H_{65}))$ (174 orthogonal pairs).*

3. Chromatic–clique gap $\chi(H_{65}) = 5 > 4 = \omega(\Gamma(H_{65}))$: *The 17-vertex core $G_{\text{core}}$ has independence number $\alpha(G_{\text{core}}) = 4$, fractional chromatic number $\chi_f(\Gamma(H_{65})) = 17/4 = 4.25 > 4$, and $\chi(H_{65}) = 5$.*

4. Set-representability: *The core $(V_{\text{core}}, \{C_1, \ldots, C_5\})$ admits exactly 24 two-valued state supports $S_1, \ldots, S_{24}$ compatible with $G_{\text{core}}$ (listed in Table II), and the resulting 48 canonical two-valued states $s_{\alpha,\varepsilon}$ on $H_{65}$ via Eq. (8) form a unital and separating collection on all 65 vertices (within a full state space of $24 \times 2^{14} = 393{,}216$ states, enumerated in Appendix B 2).*

*Proof.* (1)–(2) By inspection of Eq. (13), the five base contexts $C_1, \ldots, C_5$ satisfy $|C_a \cap C_b| \le 1$ ($C_1, C_2$ are disjoint from everything else, while $C_3 \cap C_4 = \{9\}$, $C_3 \cap C_5 = \{10\}$, $C_4 \cap C_5 = \{11\}$), and the 24 pairs $e_m = \{u_m, v_m\}$ in $E_1, \ldots, E_{24}$ are precisely the 24 orthogonal pairs between $C_1 \cup C_2$ and $B_1 \cup B_2 \cup B_3$.

To determine the triangles and 4-cliques of $G_{\text{core}}$ and $\Gamma(H_{65})$, let $e_m = \{u_m, v_m\}$ be a bridge edge with $u_m \in C_a$ ($a \in \{1, 2\}$) and $v_m \in B_k$. Because $u_m$ is non-orthogonal to the other basis $C_{3-a}$ and to $\{9, 10, 11\}$, while $v_m$ has no neighbors in $(B_1 \cup B_2 \cup B_3) \setminus B_k$ and its partner $v'_m \in B_k$ is non-orthogonal to $u_m$ (Table III), the *unique* common third neighbor of $u_m$ and $v_m$ in $G_{\text{core}}$ is the second vertex of $N(v_m) \cap C_a$. Consequently, $G_{\text{core}}$ has exactly 13 triangles not contained in $C_1, \ldots, C_5$: the 12 bridge triangles

$$
\begin{aligned}
&\{1,2,17\},\ \{1,3,15\},\ \{1,4,13\},\\
&\{2,3,12\},\ \{2,4,14\},\ \{3,4,16\},\\
&\{5,6,17\},\ \{5,7,15\},\ \{5,8,12\},\\
&\{6,7,13\},\ \{6,8,14\},\ \{7,8,16\},
\end{aligned}
$$

together with the hub triad $\{9, 10, 11\}$. Since each bridge edge has only one common third neighbor in $V_{\text{core}}$, *no 4-clique ($K_4$) in $G_{\text{core}}$ contains a bridge edge.* This has two immediate consequences:

- First, for any two distinct bridge edges $e_\ell, e_m$, the four rays $\{u_\ell, v_\ell, u_m, v_m\}$ cannot form a $K_4$ in $G_{\text{core}}$, so $L_\ell \not\perp L_m$ ($P_\ell \neq L_m$), verifying Eq. (7). Likewise, any nonzero vector in $L_m = \text{span}(v_{u_m}, v_{v_m})$ outside $\mathbb{R}v_{u_m} \cup \mathbb{R}v_{v_m}$ either has support size 3 (when $u_m \in C_1$, whereas no vector in $V_{\text{core}}$ has support size 3) or has components outside $\{-1, 0, 1\}$ up to scaling (when $u_m \in C_2$), verifying Eq. (6). The reproducible completion rule and exact integer verification in Appendix C confirm that

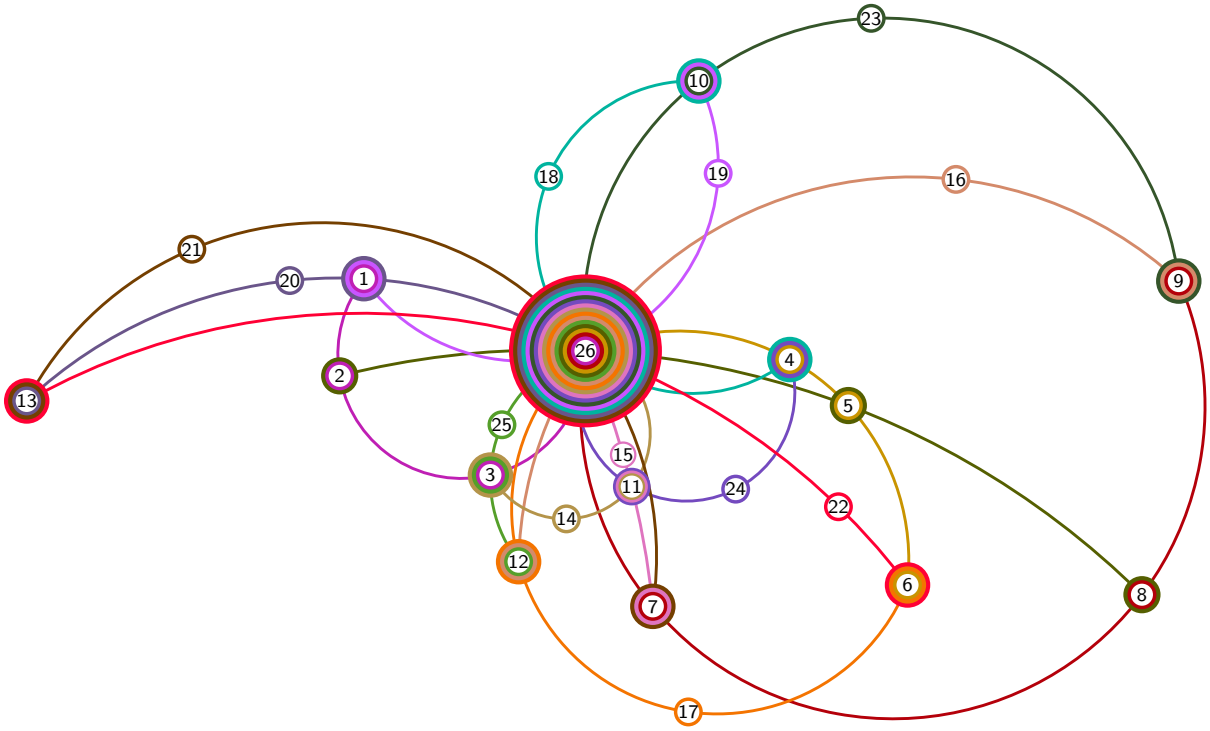



FIG. 1. Four-uniform hypergraph diagram of the 26-vertex, 16-context non-linear warm-up apex suspension $H^{(4)}_{\mathrm{YO}} = (V_{26}, E^{(4)}_{16})$ in $\mathbb{R}^4$ of Proposition 1 (26 pairwise non-collinear integer vectors $v_1, \ldots, v_{26} \in \mathbb{Z}^4$ enumerated in Eq. (A2), 16 four-element contexts $e^{(4)}_1, \ldots, e^{(4)}_{16}$ listed in Table VI, and 73 orthogonality edges). Every context $e^{(4)}_r = e_r \cup \{26\}$ $(r = 1, \ldots, 16)$ is drawn as a circular arc of a distinct color passing through its four vertices. Each vertex number is surrounded by concentric filled disks of successive radii matching the colors of its incident context arcs, with a white-filled center behind the vertex label: the 12 completion vertices $14, \ldots, 25$ have 1 colored disk, the 3 coordinate vertices $2, 5, 8$ have 2 nested disks, the 10 vertices $1, 3, 4, 6, 7, 9, 10, 11, 12, 13$ have 3 nested disks, and the common apex vertex 26 at the center is surrounded by 16 nested colored disks corresponding to all 16 contexts containing it. Unmarked arc crossings are not vertices.

TABLE II. The 24 core state supports $S_\alpha \subset V_{\mathrm{core}}$ $(\alpha = 1, \ldots, 24)$ of $H_{65}$, grouped by their hub ray $h \in \{9, 10, 11\}$ and opposite bridge ray $b \in B_1 \cup B_2 \cup B_3$.

| Hub $h$ | Bridge $b$ | Allowed $(i_1, i_2) \in C_1 \times C_2$ | Supports $S_\alpha = \{i_1, i_2, h, b\}$ |
|---|---|---|---|
| 9 | 16 | $\{1,2\} \times \{5,6\}$ | $S_1 = \{1,5,9,16\}$, $S_3 = \{1,6,9,16\}$, $S_7 = \{2,5,9,16\}$, $S_9 = \{2,6,9,16\}$ |
| 9 | 17 | $\{3,4\} \times \{7,8\}$ | $S_{15} = \{3,7,9,17\}$, $S_{17} = \{3,8,9,17\}$, $S_{21} = \{4,7,9,17\}$, $S_{23} = \{4,8,9,17\}$ |
| 10 | 14 | $\{1,3\} \times \{5,7\}$ | $S_2 = \{1,5,10,14\}$, $S_5 = \{1,7,10,14\}$, $S_{13} = \{3,5,10,14\}$, $S_{16} = \{3,7,10,14\}$ |
| 10 | 15 | $\{2,4\} \times \{6,8\}$ | $S_{10} = \{2,6,10,15\}$, $S_{11} = \{2,8,10,15\}$, $S_{19} = \{4,6,10,15\}$, $S_{24} = \{4,8,10,15\}$ |
| 11 | 12 | $\{1,4\} \times \{6,7\}$ | $S_4 = \{1,6,11,12\}$, $S_6 = \{1,7,11,12\}$, $S_{20} = \{4,6,11,12\}$, $S_{22} = \{4,7,11,12\}$ |
| 11 | 13 | $\{2,3\} \times \{5,8\}$ | $S_8 = \{2,5,11,13\}$, $S_{12} = \{2,8,11,13\}$, $S_{14} = \{3,5,11,13\}$, $S_{18} = \{3,8,11,13\}$ |

$v_i \cdot v_j = 0$ holds exactly for the $5 \times 6 + 24 \times 6 = 174$ edges of $\Gamma(H_{65})$.

- Second, deleting all 24 bridge edges from $G_{\mathrm{core}}$ leaves only the edges of $C_1, \ldots, C_5$, whose only $K_4$ cliques are $C_1, \ldots, C_5$ themselves. Since every fresh completion vertex $w \in \{18, \ldots, 65\}$ has degree 3 in $\Gamma(H_{65})$ with neighborhood equal to the remaining three vertices of its unique completion context $E_m$, the 4-cliques of $\Gamma(H_{65})$ are precisely the 29 contexts of $E_{29}$, and $\omega(\Gamma(H_{65})) = 4$ (which also follows directly from the injective orthogonal realization in $\mathbb{R}^4$).

(3) To prove $\alpha(G_{\mathrm{core}}) = 4$ by hand, let $I \subseteq V_{\mathrm{core}}$ be an independent set in $G_{\mathrm{core}}$, and write $W_9 = C_3 \cup C_4 \cup C_5 = \{9, 10, 11\} \cup B_1 \cup B_2 \cup B_3$. Since $C_1$ and $C_2$ are 4-cliques, $|I \cap C_1| \leq 1$ and $|I \cap C_2| \leq 1$. Moreover, $W_9$ is partitioned into the triangle $\{9, 10, 11\}$ and the three edges $B_1, B_2, B_3$, so $|I \cap W_9| \leq 4$; in fact, if $I \cap \{9, 10, 11\} \neq \emptyset$, say $9 \in I$, then $9 \in C_3 \cap C_4$ excludes all

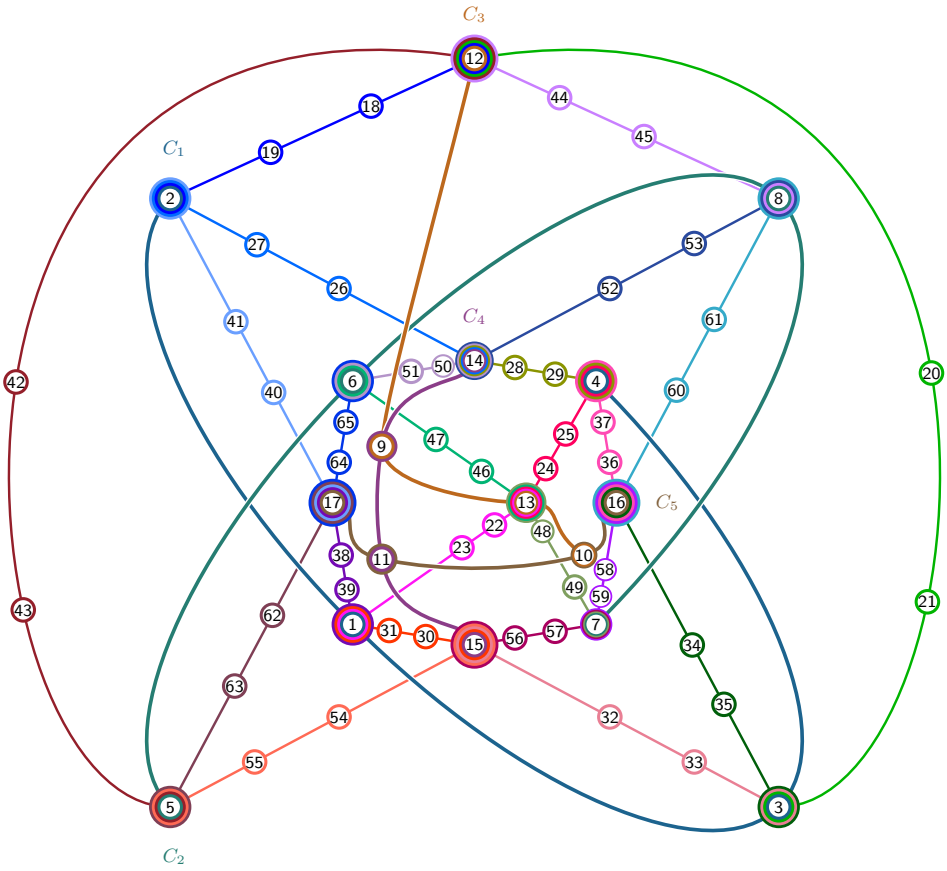



FIG. 2. Complete four-uniform hypergraph diagram of the canonical linear 4-uniform hypergraph $H_{65} = (V_{65}, E_{29})$ in $\mathbb{R}^4$ of Theorem 4 (65 pairwise non-collinear integer vectors $v_1, \ldots, v_{65} \in \mathbb{Z}^4$ enumerated in Eq. (12) and Table IV, 29 four-element contexts listed in Eq. (13) and Table V, and $29 \times \binom{4}{2} = 174$ orthogonality edges). All 29 four-element contexts have distinct colors, matched by the concentric filled disk layers of successive radii surrounding each vertex number, with a white center behind every label: the five core contexts $C_1 = \{1, 2, 3, 4\}$, $C_2 = \{5, 6, 7, 8\}$, $C_3 = \{9, 10, 12, 13\}$, $C_4 = \{9, 11, 14, 15\}$, and $C_5 = \{10, 11, 16, 17\}$ are drawn as bold smooth curves, while the 24 bridge-completion contexts $E_1, \ldots, E_{24}$ (such as the straight line $E_{21} = \{7, 16, 58, 59\}$) connect the binary bridge vertices $12, \ldots, 17$ (each surrounded by 5 nested colored disks) to the basis vertices $1, \ldots, 8$ (4 nested disks) via the degree-1 completion pairs $18, \ldots, 65$ (1 colored disk), with the hub vertices $9, 10, 11$ surrounded by 2 nested disks. Unmarked crossings are not vertices.

TABLE III. Orthogonal neighbors in $C_1 = \{1, 2, 3, 4\}$ and $C_2 = \{5, 6, 7, 8\}$ for each of the six binary bridge rays $b \in B_1 \cup B_2 \cup B_3$.

| Pair | Ray $b$ | $N(b) \cap C_1$ | $N(b) \cap C_2$ |
|---|---|---|---|
| $B_1$ | $12 = (1, 0, 0, -1)$ | $\{2, 3\}$ | $\{5, 8\}$ |
| $B_1$ | $13 = (0, 1, 1, 0)$ | $\{1, 4\}$ | $\{6, 7\}$ |
| $B_2$ | $14 = (1, 0, 1, 0)$ | $\{2, 4\}$ | $\{6, 8\}$ |
| $B_2$ | $15 = (0, 1, 0, -1)$ | $\{1, 3\}$ | $\{5, 7\}$ |
| $B_3$ | $16 = (1, 1, 0, 0)$ | $\{3, 4\}$ | $\{7, 8\}$ |
| $B_3$ | $17 = (0, 0, 1, -1)$ | $\{1, 2\}$ | $\{5, 6\}$ |

of $\{10, 11, 12, 13, 14, 15\}$ and leaves at most one vertex in $B_3 = \{16, 17\}$, so $|I \cap W_9| \leq 2$ and hence $|I| \leq 1+1+2 = 4$.

The only remaining way to have $|I| \geq 5$ is for $I \cap \{9, 10, 11\} = \emptyset$ and $|I \cap W_9| = 3$, which requires $I \cap W_9 = \{b_1, b_2, b_3\}$ with $(b_1, b_2, b_3) \in B_1 \times B_2 \times B_3$. Table III lists the orthogonal neighbors of each $b \in B_1 \cup B_2 \cup B_3$ in $C_1$ and $C_2$. Call a triple $(b_1, b_2, b_3) \in B_1 \times B_2 \times B_3$ even or odd according to the parity of the number of its entries in $\{13, 15, 17\}$, the second elements of the three ordered bridge pairs. Observe the complementary parity of the $2 + 2$ partitions of $C_1$ and $C_2$:

- For the four *even-parity* triples $(b_1, b_2, b_3) \in \{(12, 14, 16), (12, 15, 17), (13, 14, 17), (13, 15, 16)\}$, the union of their orthogonal neighbors in $C_2$ is all of $C_2 = \{5, 6, 7, 8\}$ (leaving only one non-neighbor in $C_1$ and *zero* non-neighbors in $C_2$), so $|I| \leq 3 + 1 + 0 = 4$.
- For the four *odd-parity* triples $(b_1, b_2, b_3) \in$

$\{(12,14,17),(12,15,16),(13,14,16),(13,15,17)\}$, the union of their orthogonal neighbors in $C_1$ is all of $C_1 = \{1,2,3,4\}$ (leaving *zero* non-neighbors in $C_1$ and only one non-neighbor in $C_2$), so again $|I| \le 3+0+1 = 4$.

Thus $\alpha(G_{\text{core}}) = 4$ (attained, e.g., by $S_1 = \{1,5,9,16\}$). In any proper $k$-coloring of $G_{\text{core}}$, each color class is an independent set of size at most 4, so $4k \ge |V_{\text{core}}| = 17$, forcing $k \ge \lceil 17/4 \rceil = 5$ and $\chi_f(\Gamma(H_{65})) \ge \chi_f(G_{\text{core}}) \ge 17/4 = 4.25$. Together with the explicit 5-coloring in Eq. (12), this establishes $\chi(H_{65}) = 5 > 4 = \omega(\Gamma(H_{65}))$.

For the matching fractional upper bound, the independent four-sets of $G_{\text{core}}$ consist of the 24 sets in Table II, each with one hub, one bridge vertex, and one vertex from each of $C_1, C_2$; 12 sets with two bridge vertices from different pairs and one vertex from each basis; and 8 sets with three bridge vertices and one basis vertex. These counts follow from the neighbor lists in Table III, which also show that assigning weight 1/8 to each set of the first type and 1/16 to each of the other types covers every vertex with total weight one. The total weight is $24/8 + 12/16 + 8/16 = 17/4$.

Equivalently, take two copies of each of the first 24 independent sets and one copy of each of the other 20, obtaining 68 color classes in which every core vertex occurs 16 times. For any bridge edge, its adjacent endpoints occupy 32 disjoint classes, leaving 36 classes available. Assign 16 available classes to its first completion vertex and 16 other available classes to its second. Completion vertices from different contexts are nonadjacent, so these choices are independent. This extends the coloring to all 65 vertices and proves $\chi_f(\Gamma(H_{65})) \le 68/16 = 17/4$. Together with the lower bound, this proves equality.

(4) A subset $S \subset V_{\text{core}}$ is the support of a core state compatible with $G_{\text{core}}$ if and only if $S$ is independent in $G_{\text{core}}$ and $|S \cap C_a| = 1$ for $a = 1,\dots,5$. (Without the 24 bridge-edge constraints, the five-context hypergraph $(V_{\text{core}}, \{C_1,\dots,C_5\})$ alone has $4\times4\times(3\times2+2^3) = 224$ two-valued states.) By the parity analysis in Part (3), no triple $(b_1,b_2,b_3) \in B_1\times B_2\times B_3$ can be extended to include both a vertex of $C_1$ and a vertex of $C_2$. Consequently, $S$ must select:

- exactly one hub vertex $h \in \{9,10,11\}$ (3 choices), which satisfies the two contexts among $C_3, C_4, C_5$ containing $h$;
- one binary bridge vertex $b$ from the unique remaining pair $B_k$ (2 choices), satisfying the third context among $C_3, C_4, C_5$;
- one of the two vertices $i_1 \in C_1 \setminus N(b)$ (2 choices); and
- one of the two vertices $i_2 \in C_2 \setminus N(b)$ (2 choices).

This yields precisely $3\times2\times2\times2 = 24$ graph-compatible core state supports $S_1,\dots,S_{24}$, listed in Table II.

To count the full state space $\mathcal{S}(H_{65})$, note that in the 24-edge bipartite bridge graph between $C_1 \cup C_2$ and $B_1 \cup B_2 \cup B_3$, every coordinate vertex $i_1 \in C_1$ has bridge degree 3, every Hadamard vertex $i_2 \in C_2$ has bridge degree 3, every bridge vertex $b \in B_1 \cup B_2 \cup B_3$ has bridge degree 4, and every hub vertex $h \in \{9,10,11\}$ has bridge degree 0. Because each $S_\alpha = \{i_1, i_2, h, b\}$ is independent in $G_{\text{core}}$, no bridge edge has both endpoints in $S_\alpha$; hence *every* $S_\alpha$ meets $3+3+0+4 = 10$ bridge edges and misses $24-10 = 14$ bridge edges. Each of the 14 missed completion contexts $E_m = \{u_m, v_m, w_{m,1}, w_{m,2}\}$ admits 2 independent binary extensions $(s(w_{m,1}), s(w_{m,2})) \in \{(1,0),(0,1)\}$, while the 10 hit completion contexts force $(0,0)$, giving $|\mathcal{S}(H_{65})| = 24 \times 2^{14} = 393{,}216$.

Finally, to see that the 24 edge zero-profiles $Z(e_m)$ in Eq. (9) are nonempty and pairwise distinct, partition $S_1,\dots,S_{24}$ into three blocks $\mathcal{B}_9, \mathcal{B}_{10}, \mathcal{B}_{11}$ of 8 supports each according to their hub ray $h \in \{9,10,11\}$ (the three horizontal sections of Table II). Every vertex $u \in C_1 \cup C_2$ occurs in 2 supports in each of the three blocks $\mathcal{B}_9, \mathcal{B}_{10}, \mathcal{B}_{11}$ (total weight 6, block weights $(2,2,2)$), whereas each bridge vertex $v \in B_k$ occurs in 4 supports concentrated entirely in the single block $\mathcal{B}_{h_k}$ of the hub $h_k$ opposite $B_k$ (block weights $(4,0,0)$ up to permutation). Since $u_m \perp v_m$, the indicators $\mathbf{1}_{u_m}$ and $\mathbf{1}_{v_m}$ are disjoint, so the complement of $Z(e_m)$ has indicator $f_m = \mathbf{1}_{u_m} + \mathbf{1}_{v_m}$ of weight $6+4 = 10$ (so $|Z(e_m)| = 24-10 = 14 > 0$) and block weights $(6,2,2)$ up to permutation. The unique block of weight 6 identifies $\mathcal{B}_{h_k}$ and hence $B_k$; within $\mathcal{B}_{h_k}$, $f_m$ takes value 1 on all 4 supports containing $v_m$ and on only 2 of the 4 supports containing its partner $v'_m \in B_k$, which uniquely identifies $v_m$; and subtracting $\mathbf{1}_{v_m}$ from $f_m$ recovers $\mathbf{1}_{u_m}$, which uniquely identifies $u_m \in C_1 \cup C_2$ since all 17 core vertex profiles are distinct. Thus all 24 zero-profiles $Z(e_1),\dots,Z(e_{24})$ are nonempty and pairwise distinct, and Lemma 3(3) guarantees that the 48 canonical states $s_{\alpha,\varepsilon}$ are unital and separating on all 65 vertices of $H_{65}$. □

### A. Scope of the partition representation

For either the canonical family or the full state space, the sets $A_i$ of Eq. (3) separate all named vertices and turn the listed contexts into partitions. For $H_{65}$, however, the classification of its core supports also implies

$$A_9 = A_{16} \,\dot\cup\, A_{17}, \quad A_{10} = A_{14} \,\dot\cup\, A_{15}, \quad A_{11} = A_{12} \,\dot\cup\, A_{13}. \tag{14}$$

Here $\dot\cup$ denotes disjoint union. For example, Eq. (14) gives the same set image to the ray $[v_9]$ and the plane $\mathrm{span}(v_{16}, v_{17})$, which are distinct Hilbert-space events. Thus the vertex-and-context representation is not an injective representation of all compound Hilbert events.

This distinction also appears at the probability level. Let $q = (1,1,1,-1)^T$, the missing fourth odd-parity Hadamard ray, and define $\Pi_i = v_i v_i^T / \|v_i\|^2$ and $\Pi_q =$

$qq^T/4$. Direct summation gives

$$\sum_{i=1}^{17} \Pi_i = 4I_4 + \Pi_q. \tag{15}$$

Every two-valued state has exactly four active core vertices, so every convex mixture of such states gives total core probability 4. By Eq. (15), quantum preparation along $q$, represented by the unit vector $q/2$, gives total core probability 5. The existence of separating two-valued states therefore does not imply a classical model for all quantum preparations.

## Appendix A: Explicit enumeration of unnormalized vectors, contexts, and non-collinearity certificates

For direct human and computational verification, this appendix enumerates all unnormalized integer vectors in $\mathbb{Z}^4$ and all four-element contexts for both hypergraphs studied in this paper:

1. The *canonical linear hypergraph* $H_{65} = (V_{65}, E_{29})$ (Theorem 4, Appendix A 1), which has exactly 65 **pairwise non-collinear integer vectors** (17 core vectors $v_1, \ldots, v_{17}$ and 48 degree-1 completion vectors $v_{18}, \ldots, v_{65}$), 29 **four-element contexts** (5 core contexts $C_1, \ldots, C_5$ and 24 bridge-completion contexts $E_1, \ldots, E_{24}$), and 174 **orthogonality edges** ($29 \times \binom{4}{2} = 174$).

2. The *non-linear warm-up apex suspension* $H^{(4)}_{\mathrm{YO}} = (V_{26}, E^{(4)}_{16})$ (Proposition 1, Appendix A 2), which has exactly 26 **pairwise non-collinear integer vectors** (13 Yu–Oh rays $v_1, \ldots, v_{13}$, 12 completion rays $v_{14}, \ldots, v_{25}$ in $\mathbb{R}^3 \times \{0\}$, and 1 apex ray $v_{26}$), 16 **four-element contexts**, and 73 **orthogonality edges** ($16 \times \binom{3}{2} = 48$ edges in $\mathbb{R}^3 \times \{0\}$ plus 25 apex edges).

*Canonical primitive normal form and human-verifiable non-collinearity.* Every vector $v_i = (x_{i,1}, x_{i,2}, x_{i,3}, x_{i,4})^T \in \mathbb{Z}^4$ listed below in both $H_{65}$ and $H^{(4)}_{\mathrm{YO}}$ is written in *canonical primitive projective form*:

$$\begin{gathered} \gcd\big(|x_{i,1}|, |x_{i,2}|, |x_{i,3}|, |x_{i,4}|\big) = 1, \\ \text{first nonzero entry of } v_i > 0. \end{gathered} \tag{A1}$$

Because every projective ray $[v] \in \mathbb{P}(\mathbb{Q}^4)$ possesses a *unique* representative in $\mathbb{Z}^4$ satisfying Eq. (A1), two vectors $v_i, v_j$ in canonical primitive form are collinear (proportional over $\mathbb{R}$ or $\mathbb{C}$) *if and only if they are identical* 4-*tuples* ($v_i = v_j$) and, in particular, have the same squared Euclidean norm $\|v_i\|^2 = \|v_j\|^2$.

### 1. The canonical linear 65-vertex, 29-context hypergraph $H_{65}$

Table IV enumerates all 65 unnormalized integer vectors $v_1, \ldots, v_{65} \in \mathbb{Z}^4$ of $H_{65}$, together with their squared norms $\|v_i\|^2$, their colors $c(i) \in \{0, 1, 2, 3, 4\}$ in the strong 5-coloring of Eq. (12), and the contexts in $E_{29}$ to which each vertex belongs. Table V enumerates all 29 contexts $C_1, \ldots, C_5, E_1, \ldots, E_{24} \in E_{29}$, displaying side by side both the vertex indices $\{i, j, k, \ell\}$ and the four mutually orthogonal integer vectors $\{v_i, v_j, v_k, v_\ell\} \subset \mathbb{Z}^4$.

To verify by hand that all 65 vectors in Table IV are pairwise non-collinear across all $\binom{65}{2} = 2{,}080$ pairs, observe that all 65 tuples satisfy Eq. (A1) and partition by squared norm $\|v_i\|^2$ into seven small, disjoint norm shells ($4 + 6 + 7 + 12 + 12 + 12 + 12 = 65$):

- $\|v_i\|^2 = 1$ (4 vectors): indices $\{1, 2, 3, 4\}$, the 4 standard coordinate basis vectors;
- $\|v_i\|^2 = 2$ (6 vectors): indices $\{12, \ldots, 17\}$, one vector for each of the $\binom{4}{2} = 6$ coordinate-pair supports;
- $\|v_i\|^2 = 4$ (7 vectors): indices $\{5, \ldots, 11\}$, seven distinct $\pm 1$ vectors with $x_{i,1} = +1$;
- $\|v_i\|^2 = 11$ (12 vectors): indices $\{18, 20, 22, 25, 26, 28, 30, 33, 34, 36, 39, 41\}$, each having a single zero coordinate (three each with $x_1 = 0$, $x_2 = 0$, $x_3 = 0$, $x_4 = 0$) and distinct sign/entry patterns of $\{0, \pm 1, \pm 1, \pm 3\}$;
- $\|v_i\|^2 = 18$ (12 vectors): indices $\{43, 45, 46, 49, 51, 53, 54, 56, 59, 61, 62, 65\}$, twelve distinct permutations and sign patterns of $\{\pm 1, \pm 2, \pm 2, \pm 3\}$;
- $\|v_i\|^2 = 22$ (12 vectors): indices $\{19, 21, 23, 24, 27, 29, 31, 32, 35, 37, 38, 40\}$, each having a single zero coordinate (three each with $x_1 = 0$, $x_2 = 0$, $x_3 = 0$, $x_4 = 0$) and distinct patterns of $\{0, \pm 2, \pm 3, \pm 3\}$;
- $\|v_i\|^2 = 36$ (12 vectors): indices $\{42, 44, 47, 48, 50, 52, 55, 57, 58, 60, 63, 64\}$, twelve distinct permutations and sign patterns of $\{\pm 1, \pm 1, \pm 3, \pm 5\}$.

Because no two 4-tuples within any of the seven norm shells coincide, no two of the 65 vectors are collinear; quantitatively, the maximum squared cosine over all $\binom{65}{2} = 2{,}080$ pairs is $\max_{1 \le i < j \le 65} (v_i \cdot v_j)^2 / (\|v_i\|^2 \|v_j\|^2) = 19^2/(11 \times 36) = 361/396 \approx 0.9116 < 1$, attained by $(v_{41}, v_{47})$, whose squared norms are 11 and 36 and whose inner product is 19.

### 2. The non-linear warm-up 26-vertex, 16-context apex suspension $H^{(4)}_{\mathrm{YO}}$

For completeness, the 26 unnormalized integer vectors $v_1, \ldots, v_{26} \in \mathbb{Z}^4$ of the non-linear warm-up hypergraph

TABLE IV. Complete enumeration of the 65 unnormalized integer vectors $v_i \in \mathbb{Z}^4$ of $H_{65}$, their squared norms $\|v_i\|^2$, their colors $c(i) \in \{0,1,2,3,4\}$, and the contexts in $E_{29}$ containing vertex $i$.

| $i$ | $v_i \in \mathbb{Z}^4$ | $\|v_i\|^2$ | $c(i)$ | Contexts containing $i$ | $i$ | $v_i \in \mathbb{Z}^4$ | $\|v_i\|^2$ | $c(i)$ | Contexts containing $i$ |
|---|---|---|---|---|---|---|---|---|---|
| 1 | $(1,0,0,0)$ | 1 | 0 | $C_1, E_3, E_7, E_{11}$ | 34 | $(1,-1,0,3)$ | 11 | 1 | $E_9 = \{3,16,34,35\}$ |
| 2 | $(0,1,0,0)$ | 1 | 1 | $C_1, E_1, E_5, E_{12}$ | 35 | $(3,-3,0,-2)$ | 22 | 3 | $E_9 = \{3,16,34,35\}$ |
| 3 | $(0,0,1,0)$ | 1 | 2 | $C_1, E_2, E_8, E_9$ | 36 | $(1,-1,3,0)$ | 11 | 1 | $E_{10} = \{4,16,36,37\}$ |
| 4 | $(0,0,0,1)$ | 1 | 3 | $C_1, E_4, E_6, E_{10}$ | 37 | $(3,-3,-2,0)$ | 22 | 2 | $E_{10} = \{4,16,36,37\}$ |
| 5 | $(1,1,1,1)$ | 4 | 0 | $C_2, E_{13}, E_{19}, E_{23}$ | 38 | $(0,2,3,3)$ | 22 | 1 | $E_{11} = \{1,17,38,39\}$ |
| 6 | $(1,1,-1,-1)$ | 4 | 1 | $C_2, E_{15}, E_{17}, E_{24}$ | 39 | $(0,3,-1,-1)$ | 11 | 3 | $E_{11} = \{1,17,38,39\}$ |
| 7 | $(1,-1,1,-1)$ | 4 | 2 | $C_2, E_{16}, E_{20}, E_{21}$ | 40 | $(2,0,-3,-3)$ | 22 | 0 | $E_{12} = \{2,17,40,41\}$ |
| 8 | $(1,-1,-1,1)$ | 4 | 4 | $C_2, E_{14}, E_{18}, E_{22}$ | 41 | $(3,0,1,1)$ | 11 | 3 | $E_{12} = \{2,17,40,41\}$ |
| 9 | $(1,1,-1,1)$ | 4 | 0 | $C_3, C_4$ | 42 | $(1,-5,3,1)$ | 36 | 1 | $E_{13} = \{5,12,42,43\}$ |
| 10 | $(1,-1,1,1)$ | 4 | 1 | $C_3, C_5$ | 43 | $(2,-1,-3,2)$ | 18 | 2 | $E_{13} = \{5,12,42,43\}$ |
| 11 | $(1,-1,-1,-1)$ | 4 | 3 | $C_4, C_5$ | 44 | $(1,-3,5,1)$ | 36 | 0 | $E_{14} = \{8,12,44,45\}$ |
| 12 | $(1,0,0,-1)$ | 2 | 3 | $C_3, E_1, E_2, E_{13}, E_{14}$ | 45 | $(2,3,1,2)$ | 18 | 1 | $E_{14} = \{8,12,44,45\}$ |
| 13 | $(0,1,1,0)$ | 2 | 4 | $C_3, E_3, E_4, E_{15}, E_{16}$ | 46 | $(1,-2,2,-3)$ | 18 | 0 | $E_{15} = \{6,13,46,47\}$ |
| 14 | $(1,0,1,0)$ | 2 | 2 | $C_4, E_5, E_6, E_{17}, E_{18}$ | 47 | $(5,-1,1,3)$ | 36 | 2 | $E_{15} = \{6,13,46,47\}$ |
| 15 | $(0,1,0,-1)$ | 2 | 1 | $C_4, E_7, E_8, E_{19}, E_{20}$ | 48 | $(3,-1,1,5)$ | 36 | 0 | $E_{16} = \{7,13,48,49\}$ |
| 16 | $(1,1,0,0)$ | 2 | 0 | $C_5, E_9, E_{10}, E_{21}, E_{22}$ | 49 | $(3,2,-2,-1)$ | 18 | 1 | $E_{16} = \{7,13,48,49\}$ |
| 17 | $(0,0,1,-1)$ | 2 | 2 | $C_5, E_{11}, E_{12}, E_{23}, E_{24}$ | 50 | $(1,3,-1,5)$ | 36 | 0 | $E_{17} = \{6,14,50,51\}$ |
| 18 | $(1,0,-3,1)$ | 11 | 0 | $E_1 = \{2,12,18,19\}$ | 51 | $(2,-3,-2,1)$ | 18 | 3 | $E_{17} = \{6,14,50,51\}$ |
| 19 | $(3,0,2,3)$ | 22 | 2 | $E_1 = \{2,12,18,19\}$ | 52 | $(1,5,-1,3)$ | 36 | 0 | $E_{18} = \{8,14,52,53\}$ |
| 20 | $(1,3,0,1)$ | 11 | 0 | $E_2 = \{3,12,20,21\}$ | 53 | $(2,1,-2,-3)$ | 18 | 1 | $E_{18} = \{8,14,52,53\}$ |
| 21 | $(3,-2,0,3)$ | 22 | 1 | $E_2 = \{3,12,20,21\}$ | 54 | $(3,-2,1,-2)$ | 18 | 2 | $E_{19} = \{5,15,54,55\}$ |
| 22 | $(0,1,-1,3)$ | 11 | 1 | $E_3 = \{1,13,22,23\}$ | 55 | $(3,1,-5,1)$ | 36 | 3 | $E_{19} = \{5,15,54,55\}$ |
| 23 | $(0,3,-3,-2)$ | 22 | 2 | $E_3 = \{1,13,22,23\}$ | 56 | $(1,2,3,2)$ | 18 | 0 | $E_{20} = \{7,15,56,57\}$ |
| 24 | $(2,-3,3,0)$ | 22 | 0 | $E_4 = \{4,13,24,25\}$ | 57 | $(5,1,-3,1)$ | 36 | 3 | $E_{20} = \{7,15,56,57\}$ |
| 25 | $(3,1,-1,0)$ | 11 | 1 | $E_4 = \{4,13,24,25\}$ | 58 | $(1,-1,-5,-3)$ | 36 | 1 | $E_{21} = \{7,16,58,59\}$ |
| 26 | $(1,0,-1,-3)$ | 11 | 0 | $E_5 = \{2,14,26,27\}$ | 59 | $(2,-2,-1,3)$ | 18 | 3 | $E_{21} = \{7,16,58,59\}$ |
| 27 | $(3,0,-3,2)$ | 22 | 3 | $E_5 = \{2,14,26,27\}$ | 60 | $(1,-1,-3,-5)$ | 36 | 1 | $E_{22} = \{8,16,60,61\}$ |
| 28 | $(1,-3,-1,0)$ | 11 | 0 | $E_6 = \{4,14,28,29\}$ | 61 | $(2,-2,3,-1)$ | 18 | 2 | $E_{22} = \{8,16,60,61\}$ |
| 29 | $(3,2,-3,0)$ | 22 | 1 | $E_6 = \{4,14,28,29\}$ | 62 | $(1,3,-2,-2)$ | 18 | 1 | $E_{23} = \{5,17,62,63\}$ |
| 30 | $(0,1,3,1)$ | 11 | 2 | $E_7 = \{1,15,30,31\}$ | 63 | $(5,-3,-1,-1)$ | 36 | 3 | $E_{23} = \{5,17,62,63\}$ |
| 31 | $(0,3,-2,3)$ | 22 | 3 | $E_7 = \{1,15,30,31\}$ | 64 | $(3,-5,-1,-1)$ | 36 | 0 | $E_{24} = \{6,17,64,65\}$ |
| 32 | $(2,3,0,3)$ | 22 | 0 | $E_8 = \{3,15,32,33\}$ | 65 | $(3,1,2,2)$ | 18 | 3 | $E_{24} = \{6,17,64,65\}$ |
| 33 | $(3,-1,0,-1)$ | 11 | 3 | $E_8 = \{3,15,32,33\}$ | | | | | |

$H_{\mathrm{YO}}^{(4)} = (V_{26}, E_{16}^{(4)})$ in Proposition 1 (written in canonical primitive form satisfying Eq. (A1)) are:

$$
\begin{aligned}
v_1 &= (0,1,1,0), & v_2 &= (1,0,0,0),\\
v_3 &= (0,1,-1,0), & v_4 &= (1,1,0,0),\\
v_5 &= (0,0,1,0), & v_6 &= (1,-1,0,0),\\
v_7 &= (1,0,1,0), & v_8 &= (0,1,0,0),\\
v_9 &= (1,0,-1,0), & v_{10} &= (1,-1,1,0),\\
v_{11} &= (1,-1,-1,0), & v_{12} &= (1,1,1,0),\\
v_{13} &= (1,1,-1,0), & v_{14} &= (2,1,1,0),\\
v_{15} &= (1,2,-1,0), & v_{16} &= (1,-2,1,0),\\
v_{17} &= (1,1,-2,0), & v_{18} &= (1,-1,-2,0),\\
v_{19} &= (2,1,-1,0), & v_{20} &= (2,-1,1,0),\\
v_{21} &= (1,-2,-1,0), & v_{22} &= (1,1,2,0),\\
v_{23} &= (1,2,1,0), & v_{24} &= (1,-1,2,0),\\
v_{25} &= (2,-1,-1,0), & v_{26} &= (0,0,0,1).
\end{aligned}
\tag{A2}
$$

These 26 canonical primitive vectors partition by squared norm $\|v_i\|^2$ into four disjoint norm shells ($4+6+4+12 = 26$):

- $\|v_i\|^2 = 1$ (4 vectors): $\{v_2, v_5, v_8, v_{26}\}$, the four coordinate basis vectors of $\mathbb{R}^4$;
- $\|v_i\|^2 = 2$ (6 vectors): $\{v_1, v_3, v_4, v_6, v_7, v_9\}$, the six binary rays in $\mathbb{R}^3 \times \{0\}$;
- $\|v_i\|^2 = 3$ (4 vectors): $\{v_{10}, v_{11}, v_{12}, v_{13}\}$, the four triad rays $(1, \pm 1, \pm 1, 0)^T$;
- $\|v_i\|^2 = 6$ (12 vectors): $\{v_{14}, \ldots, v_{25}\}$, all $3 \times 4 = 12$ canonical primitive permutations and sign patterns of $(\pm 2, \pm 1, \pm 1, 0)^T$.

Since all 26 canonical 4-tuples are distinct, all $\binom{26}{2} = 325$ pairs of vectors are strictly non-collinear, with maximum squared cosine $\max_{1 \le i < j \le 26} (v_i \cdot v_j)^2 / (\|v_i\|^2 \|v_j\|^2) = 4^2/(3 \times 6) = 8/9 \approx 0.8889 < 1$. Table VI lists all 16 four-element contexts of $E_{16}^{(4)}$ side by side with their four mutually orthogonal integer vectors.

TABLE V. Complete enumeration of the 29 four-element contexts of $H_{65} = (V_{65}, E_{29})$, listing for each context both its vertex indices $\{i, j, k, \ell\}$ and the four mutually orthogonal unnormalized integer vectors $\{v_i, v_j, v_k, v_\ell\} \subset \mathbb{Z}^4$.

| Context | Vertex set | $v_i$ | $v_j$ | $v_k$ | $v_\ell$ |
|---|---|---|---|---|---|
| *Five core 4-contexts $C_1, \ldots, C_5$ on $V_{\text{core}} = \{1, \ldots, 17\}$:* | | | | | |
| $C_1$ | $\{1, 2, 3, 4\}$ | $v_1 = (1, 0, 0, 0)$ | $v_2 = (0, 1, 0, 0)$ | $v_3 = (0, 0, 1, 0)$ | $v_4 = (0, 0, 0, 1)$ |
| $C_2$ | $\{5, 6, 7, 8\}$ | $v_5 = (1, 1, 1, 1)$ | $v_6 = (1, 1, -1, -1)$ | $v_7 = (1, -1, 1, -1)$ | $v_8 = (1, -1, -1, 1)$ |
| $C_3$ | $\{9, 10, 12, 13\}$ | $v_9 = (1, 1, -1, 1)$ | $v_{10} = (1, -1, 1, 1)$ | $v_{12} = (1, 0, 0, -1)$ | $v_{13} = (0, 1, 1, 0)$ |
| $C_4$ | $\{9, 11, 14, 15\}$ | $v_9 = (1, 1, -1, 1)$ | $v_{11} = (1, -1, -1, -1)$ | $v_{14} = (1, 0, 1, 0)$ | $v_{15} = (0, 1, 0, -1)$ |
| $C_5$ | $\{10, 11, 16, 17\}$ | $v_{10} = (1, -1, 1, 1)$ | $v_{11} = (1, -1, -1, -1)$ | $v_{16} = (1, 1, 0, 0)$ | $v_{17} = (0, 0, 1, -1)$ |
| *Twelve coordinate-to-bridge completion contexts $E_1, \ldots, E_{12}$ (between $C_1$ and $B_1 \cup B_2 \cup B_3$):* | | | | | |
| $E_1$ | $\{2, 12, 18, 19\}$ | $v_2 = (0, 1, 0, 0)$ | $v_{12} = (1, 0, 0, -1)$ | $v_{18} = (1, 0, -3, 1)$ | $v_{19} = (3, 0, 2, 3)$ |
| $E_2$ | $\{3, 12, 20, 21\}$ | $v_3 = (0, 0, 1, 0)$ | $v_{12} = (1, 0, 0, -1)$ | $v_{20} = (1, 3, 0, 1)$ | $v_{21} = (3, -2, 0, 3)$ |
| $E_3$ | $\{1, 13, 22, 23\}$ | $v_1 = (1, 0, 0, 0)$ | $v_{13} = (0, 1, 1, 0)$ | $v_{22} = (0, 1, -1, 3)$ | $v_{23} = (0, 3, -3, -2)$ |
| $E_4$ | $\{4, 13, 24, 25\}$ | $v_4 = (0, 0, 0, 1)$ | $v_{13} = (0, 1, 1, 0)$ | $v_{24} = (2, -3, 3, 0)$ | $v_{25} = (3, 1, -1, 0)$ |
| $E_5$ | $\{2, 14, 26, 27\}$ | $v_2 = (0, 1, 0, 0)$ | $v_{14} = (1, 0, 1, 0)$ | $v_{26} = (1, 0, -1, -3)$ | $v_{27} = (3, 0, -3, 2)$ |
| $E_6$ | $\{4, 14, 28, 29\}$ | $v_4 = (0, 0, 0, 1)$ | $v_{14} = (1, 0, 1, 0)$ | $v_{28} = (1, -3, -1, 0)$ | $v_{29} = (3, 2, -3, 0)$ |
| $E_7$ | $\{1, 15, 30, 31\}$ | $v_1 = (1, 0, 0, 0)$ | $v_{15} = (0, 1, 0, -1)$ | $v_{30} = (0, 1, 3, 1)$ | $v_{31} = (0, 3, -2, 3)$ |
| $E_8$ | $\{3, 15, 32, 33\}$ | $v_3 = (0, 0, 1, 0)$ | $v_{15} = (0, 1, 0, -1)$ | $v_{32} = (2, 3, 0, 3)$ | $v_{33} = (3, -1, 0, -1)$ |
| $E_9$ | $\{3, 16, 34, 35\}$ | $v_3 = (0, 0, 1, 0)$ | $v_{16} = (1, 1, 0, 0)$ | $v_{34} = (1, -1, 0, 3)$ | $v_{35} = (3, -3, 0, -2)$ |
| $E_{10}$ | $\{4, 16, 36, 37\}$ | $v_4 = (0, 0, 0, 1)$ | $v_{16} = (1, 1, 0, 0)$ | $v_{36} = (1, -1, 3, 0)$ | $v_{37} = (3, -3, -2, 0)$ |
| $E_{11}$ | $\{1, 17, 38, 39\}$ | $v_1 = (1, 0, 0, 0)$ | $v_{17} = (0, 0, 1, -1)$ | $v_{38} = (0, 2, 3, 3)$ | $v_{39} = (0, 3, -1, -1)$ |
| $E_{12}$ | $\{2, 17, 40, 41\}$ | $v_2 = (0, 1, 0, 0)$ | $v_{17} = (0, 0, 1, -1)$ | $v_{40} = (2, 0, -3, -3)$ | $v_{41} = (3, 0, 1, 1)$ |
| *Twelve Hadamard-to-bridge completion contexts $E_{13}, \ldots, E_{24}$ (between $C_2$ and $B_1 \cup B_2 \cup B_3$):* | | | | | |
| $E_{13}$ | $\{5, 12, 42, 43\}$ | $v_5 = (1, 1, 1, 1)$ | $v_{12} = (1, 0, 0, -1)$ | $v_{42} = (1, -5, 3, 1)$ | $v_{43} = (2, -1, -3, 2)$ |
| $E_{14}$ | $\{8, 12, 44, 45\}$ | $v_8 = (1, -1, -1, 1)$ | $v_{12} = (1, 0, 0, -1)$ | $v_{44} = (1, -3, 5, 1)$ | $v_{45} = (2, 3, 1, 2)$ |
| $E_{15}$ | $\{6, 13, 46, 47\}$ | $v_6 = (1, 1, -1, -1)$ | $v_{13} = (0, 1, 1, 0)$ | $v_{46} = (1, -2, 2, -3)$ | $v_{47} = (5, -1, 1, 3)$ |
| $E_{16}$ | $\{7, 13, 48, 49\}$ | $v_7 = (1, -1, 1, -1)$ | $v_{13} = (0, 1, 1, 0)$ | $v_{48} = (3, -1, 1, 5)$ | $v_{49} = (3, 2, -2, -1)$ |
| $E_{17}$ | $\{6, 14, 50, 51\}$ | $v_6 = (1, 1, -1, -1)$ | $v_{14} = (1, 0, 1, 0)$ | $v_{50} = (1, 3, -1, 5)$ | $v_{51} = (2, -3, -2, 1)$ |
| $E_{18}$ | $\{8, 14, 52, 53\}$ | $v_8 = (1, -1, -1, 1)$ | $v_{14} = (1, 0, 1, 0)$ | $v_{52} = (1, 5, -1, 3)$ | $v_{53} = (2, 1, -2, -3)$ |
| $E_{19}$ | $\{5, 15, 54, 55\}$ | $v_5 = (1, 1, 1, 1)$ | $v_{15} = (0, 1, 0, -1)$ | $v_{54} = (3, -2, 1, -2)$ | $v_{55} = (3, 1, -5, 1)$ |
| $E_{20}$ | $\{7, 15, 56, 57\}$ | $v_7 = (1, -1, 1, -1)$ | $v_{15} = (0, 1, 0, -1)$ | $v_{56} = (1, 2, 3, 2)$ | $v_{57} = (5, 1, -3, 1)$ |
| $E_{21}$ | $\{7, 16, 58, 59\}$ | $v_7 = (1, -1, 1, -1)$ | $v_{16} = (1, 1, 0, 0)$ | $v_{58} = (1, -1, -5, -3)$ | $v_{59} = (2, -2, -1, 3)$ |
| $E_{22}$ | $\{8, 16, 60, 61\}$ | $v_8 = (1, -1, -1, 1)$ | $v_{16} = (1, 1, 0, 0)$ | $v_{60} = (1, -1, -3, -5)$ | $v_{61} = (2, -2, 3, -1)$ |
| $E_{23}$ | $\{5, 17, 62, 63\}$ | $v_5 = (1, 1, 1, 1)$ | $v_{17} = (0, 0, 1, -1)$ | $v_{62} = (1, 3, -2, -2)$ | $v_{63} = (5, -3, -1, -1)$ |
| $E_{24}$ | $\{6, 17, 64, 65\}$ | $v_6 = (1, 1, -1, -1)$ | $v_{17} = (0, 0, 1, -1)$ | $v_{64} = (3, -5, -1, -1)$ | $v_{65} = (3, 1, 2, 2)$ |

## Appendix B: Explicit enumeration of two-valued states

This appendix provides the complete enumeration of all two-valued states on both hypergraphs studied in the main text: the 25 two-valued states of the non-linear warm-up hypergraph $H_{\text{YO}}^{(4)}$ (Appendix B 1), and the complete branch-by-branch parametrization of all $24 \times 2^{14} = 393{,}216$ two-valued states—together with the 48 canonical unital and separating states $s_{\alpha,\varepsilon}$ and the 65 vertex state-profiles—of the canonical linear hypergraph $H_{65}$ (Appendix B 2).

### 1. Complete enumeration of the 25 two-valued states on $H_{\text{YO}}^{(4)}$

Table VII lists all 25 two-valued states $s_1, \ldots, s_{25} : V_{26} \to \{0, 1\}$ on $H_{\text{YO}}^{(4)} = (V_{26}, E_{16}^{(4)})$. States $s_1, \ldots, s_{24}$ have $s_r(26) = 0$ and correspond to the 24 two-valued states of the completed Yu–Oh hypergraph $H_{\text{YO}}^{+}$ in $\mathbb{R}^3$ (grouped into 12 pairs of states sharing the same support on the three coordinate contexts $\{1, 2, 3\}$, $\{4, 5, 6\}$, $\{7, 8, 9\}$), while $s_{25}$ is the unique apex state with support $\{26\}$. Table VIII displays the transposed $26 \times 25$ vertex-by-state binary matrix, listing for each vertex $i \in \{1, \ldots, 26\}$ its state weight $\sum_{r=1}^{25} s_r(i) \in \{1, 3, 8, 13\}$, the set of states in which $s_r(i) = 1$, and its 25-bit profile $(s_1(i), \ldots, s_{25}(i))$. Since all 26 row weights are positive and all 26 binary rows are distinct, $\mathcal{S}(H_{\text{YO}}^{(4)})$ is unital and separating.

### 2. Complete parametrization of all 393,216 two-valued states and the 48 canonical separating states on $H_{65}$

Every two-valued state $s \in \mathcal{S}(H_{65})$ on the canonical linear hypergraph $H_{65} = (V_{65}, E_{29})$ is uniquely indexed by a pair $(\alpha, \varepsilon)$ consisting of:

1. a branch index $\alpha \in \{1, \ldots, 24\}$ selecting one of the 24 four-element core state supports $S_\alpha \subset V_{\text{core}} = \{1, \ldots, 17\}$; and

TABLE VI. Complete enumeration of the 16 four-element contexts of the non-linear warm-up apex suspension $H_{\mathrm{YO}}^{(4)} = (V_{26}, E_{16}^{(4)})$, listing for each context both its vertex indices $\{i, j, k, 26\}$ and the four mutually orthogonal unnormalized integer vectors $\{v_i, v_j, v_k, v_{26}\} \subset \mathbb{Z}^4$.

| Context | Vertex set | $v_i$ | $v_j$ | $v_k$ | $v_{26}$ |
|---|---|---|---|---|---|
| $e_1^{(4)}$ | $\{1, 2, 3, 26\}$ | $v_1 = (0, 1, 1, 0)$ | $v_2 = (1, 0, 0, 0)$ | $v_3 = (0, 1, -1, 0)$ | $v_{26} = (0, 0, 0, 1)$ |
| $e_2^{(4)}$ | $\{7, 8, 9, 26\}$ | $v_7 = (1, 0, 1, 0)$ | $v_8 = (0, 1, 0, 0)$ | $v_9 = (1, 0, -1, 0)$ | $v_{26} = (0, 0, 0, 1)$ |
| $e_3^{(4)}$ | $\{4, 5, 6, 26\}$ | $v_4 = (1, 1, 0, 0)$ | $v_5 = (0, 0, 1, 0)$ | $v_6 = (1, -1, 0, 0)$ | $v_{26} = (0, 0, 0, 1)$ |
| $e_4^{(4)}$ | $\{2, 5, 8, 26\}$ | $v_2 = (1, 0, 0, 0)$ | $v_5 = (0, 0, 1, 0)$ | $v_8 = (0, 1, 0, 0)$ | $v_{26} = (0, 0, 0, 1)$ |
| $e_5^{(4)}$ | $\{3, 12, 25, 26\}$ | $v_3 = (0, 1, -1, 0)$ | $v_{12} = (1, 1, 1, 0)$ | $v_{25} = (2, -1, -1, 0)$ | $v_{26} = (0, 0, 0, 1)$ |
| $e_6^{(4)}$ | $\{9, 12, 16, 26\}$ | $v_9 = (1, 0, -1, 0)$ | $v_{12} = (1, 1, 1, 0)$ | $v_{16} = (1, -2, 1, 0)$ | $v_{26} = (0, 0, 0, 1)$ |
| $e_7^{(4)}$ | $\{6, 12, 17, 26\}$ | $v_6 = (1, -1, 0, 0)$ | $v_{12} = (1, 1, 1, 0)$ | $v_{17} = (1, 1, -2, 0)$ | $v_{26} = (0, 0, 0, 1)$ |
| $e_8^{(4)}$ | $\{3, 11, 14, 26\}$ | $v_3 = (0, 1, -1, 0)$ | $v_{11} = (1, -1, -1, 0)$ | $v_{14} = (2, 1, 1, 0)$ | $v_{26} = (0, 0, 0, 1)$ |
| $e_9^{(4)}$ | $\{7, 11, 15, 26\}$ | $v_7 = (1, 0, 1, 0)$ | $v_{11} = (1, -1, -1, 0)$ | $v_{15} = (1, 2, -1, 0)$ | $v_{26} = (0, 0, 0, 1)$ |
| $e_{10}^{(4)}$ | $\{4, 11, 24, 26\}$ | $v_4 = (1, 1, 0, 0)$ | $v_{11} = (1, -1, -1, 0)$ | $v_{24} = (1, -1, 2, 0)$ | $v_{26} = (0, 0, 0, 1)$ |
| $e_{11}^{(4)}$ | $\{9, 10, 23, 26\}$ | $v_9 = (1, 0, -1, 0)$ | $v_{10} = (1, -1, 1, 0)$ | $v_{23} = (1, 2, 1, 0)$ | $v_{26} = (0, 0, 0, 1)$ |
| $e_{12}^{(4)}$ | $\{1, 10, 19, 26\}$ | $v_1 = (0, 1, 1, 0)$ | $v_{10} = (1, -1, 1, 0)$ | $v_{19} = (2, 1, -1, 0)$ | $v_{26} = (0, 0, 0, 1)$ |
| $e_{13}^{(4)}$ | $\{4, 10, 18, 26\}$ | $v_4 = (1, 1, 0, 0)$ | $v_{10} = (1, -1, 1, 0)$ | $v_{18} = (1, -1, -2, 0)$ | $v_{26} = (0, 0, 0, 1)$ |
| $e_{14}^{(4)}$ | $\{1, 13, 20, 26\}$ | $v_1 = (0, 1, 1, 0)$ | $v_{13} = (1, 1, -1, 0)$ | $v_{20} = (2, -1, 1, 0)$ | $v_{26} = (0, 0, 0, 1)$ |
| $e_{15}^{(4)}$ | $\{7, 13, 21, 26\}$ | $v_7 = (1, 0, 1, 0)$ | $v_{13} = (1, 1, -1, 0)$ | $v_{21} = (1, -2, -1, 0)$ | $v_{26} = (0, 0, 0, 1)$ |
| $e_{16}^{(4)}$ | $\{6, 13, 22, 26\}$ | $v_6 = (1, -1, 0, 0)$ | $v_{13} = (1, 1, -1, 0)$ | $v_{22} = (1, 1, 2, 0)$ | $v_{26} = (0, 0, 0, 1)$ |

TABLE VII. Complete enumeration of the 25 two-valued states $s_1, \ldots, s_{25} : V_{26} \to \{0, 1\}$ on the non-linear warm-up hypergraph $H_{\mathrm{YO}}^{(4)} = (V_{26}, E_{16}^{(4)})$, showing for each state its support on the 13-ray Yu–Oh core $\{1, \ldots, 13\}$, its forced support on $\{14, \ldots, 26\}$, its total support size $|S_r|$, and its 26-bit binary vector $(s_r(1 \ldots 13)\, s_r(14 \ldots 25)\, s_r(26))$.

| State | Core support $S_r \cap \{1, \ldots, 13\}$ | Completion/apex support $S_r \cap \{14, \ldots, 26\}$ | $\lvert S_r \rvert$ | Binary vector $(s_r(1 \ldots 13)\, s_r(14 \ldots 25)\, s_r(26))$ |
|---|---|---|---|---|
| $s_1$ | $\{1, 4, 8, 12\}$ | $\{14, 15, 21, 22, 23\}$ | 9 | 1001000100010 110000011100 0 |
| $s_2$ | $\{1, 4, 8\}$ | $\{14, 15, 16, 17, 21, 22, 23, 25\}$ | 11 | 1001000100000 111100011101 0 |
| $s_3$ | $\{1, 5, 7, 12\}$ | $\{14, 18, 22, 23, 24\}$ | 9 | 1000101000010 100010011110 0 |
| $s_4$ | $\{1, 5, 7\}$ | $\{14, 16, 17, 18, 22, 23, 24, 25\}$ | 11 | 1000101000000 101110011111 0 |
| $s_5$ | $\{1, 5, 9, 11\}$ | $\{17, 18, 21, 22, 25\}$ | 9 | 1000100010100 000110011001 0 |
| $s_6$ | $\{1, 5, 9\}$ | $\{14, 15, 17, 18, 21, 22, 24, 25\}$ | 11 | 1000100010000 110110011011 0 |
| $s_7$ | $\{1, 6, 8, 11\}$ | $\{16, 18, 21, 23, 25\}$ | 9 | 1000010100100 001010010101 0 |
| $s_8$ | $\{1, 6, 8\}$ | $\{14, 15, 16, 18, 21, 23, 24, 25\}$ | 11 | 1000010100000 111010010111 0 |
| $s_9$ | $\{2, 4, 7, 12\}$ | $\{14, 19, 20, 22, 23\}$ | 9 | 0101001000010 100001101100 0 |
| $s_{10}$ | $\{2, 4, 7\}$ | $\{14, 16, 17, 19, 20, 22, 23, 25\}$ | 11 | 0101001000000 101101101101 0 |
| $s_{11}$ | $\{2, 4, 9, 13\}$ | $\{14, 15, 17, 19, 25\}$ | 9 | 0101000010001 110101000001 0 |
| $s_{12}$ | $\{2, 4, 9\}$ | $\{14, 15, 17, 19, 20, 21, 22, 25\}$ | 11 | 0101000010000 110101111001 0 |
| $s_{13}$ | $\{2, 6, 7, 10\}$ | $\{14, 16, 20, 24, 25\}$ | 9 | 0100011001000 101000100011 0 |
| $s_{14}$ | $\{2, 6, 7\}$ | $\{14, 16, 18, 19, 20, 23, 24, 25\}$ | 11 | 0100011000000 101011100111 0 |
| $s_{15}$ | $\{2, 6, 9, 11\}$ | $\{18, 19, 20, 21, 25\}$ | 9 | 0100010010100 000011110001 0 |
| $s_{16}$ | $\{2, 6, 9\}$ | $\{14, 15, 18, 19, 20, 21, 24, 25\}$ | 11 | 0100010010000 110011110011 0 |
| $s_{17}$ | $\{3, 4, 8, 13\}$ | $\{15, 16, 17, 19, 23\}$ | 9 | 0011000100001 011101000100 0 |
| $s_{18}$ | $\{3, 4, 8\}$ | $\{15, 16, 17, 19, 20, 21, 22, 23\}$ | 11 | 0011000100000 011101111100 0 |
| $s_{19}$ | $\{3, 5, 7, 10\}$ | $\{16, 17, 20, 22, 24\}$ | 9 | 0010101001000 001100101010 0 |
| $s_{20}$ | $\{3, 5, 7\}$ | $\{16, 17, 18, 19, 20, 22, 23, 24\}$ | 11 | 0010101000000 001111101110 0 |
| $s_{21}$ | $\{3, 5, 9, 13\}$ | $\{15, 17, 18, 19, 24\}$ | 9 | 0010100010001 010111000010 0 |
| $s_{22}$ | $\{3, 5, 9\}$ | $\{15, 17, 18, 19, 20, 21, 22, 24\}$ | 11 | 0010100010000 010111111010 0 |
| $s_{23}$ | $\{3, 6, 8, 10\}$ | $\{15, 16, 20, 21, 24\}$ | 9 | 0010010101000 011000110010 0 |
| $s_{24}$ | $\{3, 6, 8\}$ | $\{15, 16, 18, 19, 20, 21, 23, 24\}$ | 11 | 0010010100000 011011110110 0 |
| $s_{25}$ | $\emptyset$ | $\{26\}$ | 1 | 0000000000000 000000000000 1 |

2. a 14-bit binary vector $\boldsymbol{\varepsilon} = (\varepsilon_m)_{m \in \mathcal{F}_\alpha} \in \{0, 1\}^{14}$ indexed by the 14 *free completion contexts*

$$\mathcal{F}_\alpha = \big\{ m \in \{1, \ldots, 24\} : S_\alpha \cap \{u_m, v_m\} = \emptyset \big\},$$

whose complement is the set of 10 *hit completion contexts* $\mathcal{H}_\alpha = \{1, \ldots, 24\} \setminus \mathcal{F}_\alpha$.

Explicitly, the state $s_{\alpha,\boldsymbol{\varepsilon}} : V_{65} \to \{0, 1\}$ takes values

$$\begin{aligned} s_{\alpha,\boldsymbol{\varepsilon}}(i) &= \mathbf{1}_{i \in S_\alpha} \qquad (1 \le i \le 17), \\ \big(s_{\alpha,\boldsymbol{\varepsilon}}(2m+16),\, s_{\alpha,\boldsymbol{\varepsilon}}(2m+17)\big) &= \begin{cases} (0, 0), & m \in \mathcal{H}_\alpha, \\ (1 - \varepsilon_m,\, \varepsilon_m), & m \in \mathcal{F}_\alpha. \end{cases} \end{aligned} \tag{B1}$$

TABLE VIII. Vertex state profiles across the 25 two-valued states $s_1, \ldots, s_{25}$ of $H_{\mathrm{YO}}^{(4)}$, showing for each vertex $i \in \{1, \ldots, 26\}$ its state weight $\sum_{r=1}^{25} s_r(i)$, the active state indices $\{r : s_r(i) = 1\}$, and the 25-bit profile $(s_{1\ldots12}(i)\, s_{13\ldots24}(i)\, s_{25}(i))$. All 26 rows are nonzero and pairwise distinct.

| Vertex $i$ | Weight $\sum_r s_r(i)$ | Active states $\{r \in \{1, \ldots, 25\} : s_r(i) = 1\}$ | 25-bit profile $(s_{1\ldots12}(i)\, s_{13\ldots24}(i)\, s_{25}(i))$ |
|---|---|---|---|
| 1 | 8 | $\{1, 2, 3, 4, 5, 6, 7, 8\}$ | 111111110000 000000000000 0 |
| 2 | 8 | $\{9, 10, 11, 12, 13, 14, 15, 16\}$ | 000000001111 111100000000 0 |
| 3 | 8 | $\{17, 18, 19, 20, 21, 22, 23, 24\}$ | 000000000000 000011111111 0 |
| 4 | 8 | $\{1, 2, 9, 10, 11, 12, 17, 18\}$ | 110000001111 000011000000 0 |
| 5 | 8 | $\{3, 4, 5, 6, 19, 20, 21, 22\}$ | 001111000000 000000111100 0 |
| 6 | 8 | $\{7, 8, 13, 14, 15, 16, 23, 24\}$ | 000000110000 111100000011 0 |
| 7 | 8 | $\{3, 4, 9, 10, 13, 14, 19, 20\}$ | 001100001100 110000110000 0 |
| 8 | 8 | $\{1, 2, 7, 8, 17, 18, 23, 24\}$ | 110000110000 000011000011 0 |
| 9 | 8 | $\{5, 6, 11, 12, 15, 16, 21, 22\}$ | 000011000011 001100001100 0 |
| 10 | 3 | $\{13, 19, 23\}$ | 000000000000 100000100010 0 |
| 11 | 3 | $\{5, 7, 15\}$ | 000010100000 001000000000 0 |
| 12 | 3 | $\{1, 3, 9\}$ | 101000001000 000000000000 0 |
| 13 | 3 | $\{11, 17, 21\}$ | 000000000010 000010001000 0 |
| 14 | 13 | $\{1, 2, 3, 4, 6, 8, 9, 10, 11, 12, 13, 14, 16\}$ | 111101011111 110100000000 0 |
| 15 | 13 | $\{1, 2, 6, 8, 11, 12, 16, 17, 18, 21, 22, 23, 24\}$ | 110001010011 000111001111 0 |
| 16 | 13 | $\{2, 4, 7, 8, 10, 13, 14, 17, 18, 19, 20, 23, 24\}$ | 010100110100 110011110011 0 |
| 17 | 13 | $\{2, 4, 5, 6, 10, 11, 12, 17, 18, 19, 20, 21, 22\}$ | 010111000111 000011111100 0 |
| 18 | 13 | $\{3, 4, 5, 6, 7, 8, 14, 15, 16, 20, 21, 22, 24\}$ | 001111110000 011100011101 0 |
| 19 | 13 | $\{9, 10, 11, 12, 14, 15, 16, 17, 18, 20, 21, 22, 24\}$ | 000000001111 011111011101 0 |
| 20 | 13 | $\{9, 10, 12, 13, 14, 15, 16, 18, 19, 20, 22, 23, 24\}$ | 000000001101 111101110111 0 |
| 21 | 13 | $\{1, 2, 5, 6, 7, 8, 12, 15, 16, 18, 22, 23, 24\}$ | 110011110001 001101000111 0 |
| 22 | 13 | $\{1, 2, 3, 4, 5, 6, 9, 10, 12, 18, 19, 20, 22\}$ | 111111001101 000001110100 0 |
| 23 | 13 | $\{1, 2, 3, 4, 7, 8, 9, 10, 14, 17, 18, 20, 24\}$ | 111100111100 010011010001 0 |
| 24 | 13 | $\{3, 4, 6, 8, 13, 14, 16, 19, 20, 21, 22, 23, 24\}$ | 001101010000 110100111111 0 |
| 25 | 13 | $\{2, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16\}$ | 010111110111 111100000000 0 |
| 26 | 1 | $\{25\}$ | 000000000000 000000000000 1 |

Because $|\mathcal{F}_\alpha| = 14$ for every $\alpha \in \{1, \ldots, 24\}$, each branch $\alpha$ yields $2^{14} = 16{,}384$ states $s_{\alpha,\boldsymbol{\varepsilon}}$, giving $|\mathcal{S}(H_{65})| = 24 \times 2^{14} = 393{,}216$ two-valued states in total. Setting all 14 free bits equal to $\varepsilon \in \{0, 1\}$ ($\varepsilon_m = \varepsilon$ for all $m \in \mathcal{F}_\alpha$) yields the 48 *canonical states* $s_{\alpha,0}, s_{\alpha,1}$ of Eq. (8), which have support size $|S_\alpha| + 14 = 18$:

$$\begin{aligned} \mathrm{supp}(s_{\alpha,0}) &= S_\alpha \cup \{2m + 16 : m \in \mathcal{F}_\alpha\}, \\ \mathrm{supp}(s_{\alpha,1}) &= S_\alpha \cup \{2m + 17 : m \in \mathcal{F}_\alpha\}. \end{aligned} \tag{B2}$$

Table IX enumerates all 24 branches $\alpha = 1, \ldots, 24$, specifying $S_\alpha$, the 10 hit context indices $\mathcal{H}_\alpha \subset \{1, \ldots, 24\}$, the 14 free context indices $\mathcal{F}_\alpha \subset \{1, \ldots, 24\}$, and the 24-bit free-context indicator string. Table X lists the resulting state profiles and weights for all 65 vertices of $H_{65}$ across both the 48 canonical states $\{s_{\alpha,\varepsilon}\}$ and the full space of 393,216 states: across the 48 canonical states $(s_{1,0}, s_{1,1}, \ldots, s_{24,0}, s_{24,1})$, each '1' in row $i$ of Table X expands to '(1,1)' for core vertices $i \in \{1, \ldots, 17\}$, to '(1,0)' for first completion vertices $w_{m,1} = 2m{+}16$, and to '(0,1)' for second completion vertices $w_{m,2} = 2m + 17$ (and each '0' expands to '(0,0)'), making all 65 canonical 48-bit vertex profiles nonzero and pairwise distinct by inspection.

## Appendix C: Reproducible completion rule and exact verification

The coordinates in Eq. (12) can be reproduced by the following finite selection rule. Normalize an integer vector by dividing by the greatest common divisor of its entries and choosing the sign so that its first nonzero entry is positive. Form the set of normalized nonzero vectors obtained from $\{-3, -2, \ldots, 3\}^4$, ordered by squared norm, then maximum absolute entry, then lexicographically. Process bridge edges in the order $e_{13}, \ldots, e_{24}, e_1, \ldots, e_{12}$. For each pair, choose $\eta_A$ to be the first vector in this ordered pool orthogonal to its endpoints, and $\eta_B$ the first such vector also orthogonal to $\eta_A$.

For all $(a, b) \in \{-7, -6, \ldots, 7\}^2 \setminus \{(0, 0)\}$, form the pair in Eq. (10) and normalize both vectors. Discard pairs that repeat an already placed ray or have an orthogonality to any already placed vertex other than the two endpoints. Among the remaining ordered pairs choose the lexicographically smallest score

$$\left( \max_{j,k} |(w_{m,j})_k|,\ \|w_{m,1}\|^2 + \|w_{m,2}\|^2,\ w_{m,1}, w_{m,2} \right). \tag{C1}$$

In Eq. (C1), the order on the last two entries is the usual lexicographic order on their integer coordinates. This

TABLE IX. Complete branch-by-branch parametrization of the $24 \times 2^{14} = 393{,}216$ two-valued states $s_{\alpha,\boldsymbol{\varepsilon}}$ and the 48 canonical states $s_{\alpha,0}, s_{\alpha,1}$ on $H_{65}$ via Eqs. (B1)–(B2). For each $\alpha \in \{1, \ldots, 24\}$, the table lists the 4-element core support $S_\alpha$, the 10 hit completion context indices $m \in \mathcal{H}_\alpha$ where $(s(2m+16), s(2m+17)) = (0,0)$ is forced on $E_m$, the 14 free completion context indices $m \in \mathcal{F}_\alpha$ where $(s(2m+16), s(2m+17)) = (1-\varepsilon_m, \varepsilon_m)$ with $\varepsilon_m \in \{0,1\}$ on $E_m$, and the 24-bit indicator string $(\mathbf{1}_{1\in\mathcal{F}_\alpha} \ldots \mathbf{1}_{12\in\mathcal{F}_\alpha}\, \mathbf{1}_{13\in\mathcal{F}_\alpha} \ldots \mathbf{1}_{24\in\mathcal{F}_\alpha})$.

| Branch | Core support $S_\alpha$ | Hit contexts $\mathcal{H}_\alpha$ (forced $(0,0)$) | Free contexts $\mathcal{F}_\alpha$ ($2^{14}$ choices) | Free indicator $m = 1 \ldots 24$ |
|---|---|---|---|---|
| $S_1$ | $\{1,5,9,16\}$ | $\{3,7,9,10,11,13,19,21,22,23\}$ | $\{1,2,4,5,6,8,12,14,15,16,17,18,20,24\}$ | 110111010001 011111010001 |
| $S_2$ | $\{1,5,10,14\}$ | $\{3,5,6,7,11,13,17,18,19,23\}$ | $\{1,2,4,8,9,10,12,14,15,16,20,21,22,24\}$ | 110100011101 011100011101 |
| $S_3$ | $\{1,6,9,16\}$ | $\{3,7,9,10,11,15,17,21,22,24\}$ | $\{1,2,4,5,6,8,12,13,14,16,18,19,20,23\}$ | 110111010001 110101110010 |
| $S_4$ | $\{1,6,11,12\}$ | $\{1,2,3,7,11,13,14,15,17,24\}$ | $\{4,5,6,8,9,10,12,16,18,19,20,21,22,23\}$ | 000111011101 000101111110 |
| $S_5$ | $\{1,7,10,14\}$ | $\{3,5,6,7,11,16,17,18,20,21\}$ | $\{1,2,4,8,9,10,12,13,14,15,19,22,23,24\}$ | 110100011101 111000100111 |
| $S_6$ | $\{1,7,11,12\}$ | $\{1,2,3,7,11,13,14,16,20,21\}$ | $\{4,5,6,8,9,10,12,15,17,18,19,22,23,24\}$ | 000111011101 001011100111 |
| $S_7$ | $\{2,5,9,16\}$ | $\{1,5,9,10,12,13,19,21,22,23\}$ | $\{2,3,4,6,7,8,11,14,15,16,17,18,20,24\}$ | 011101110010 011111010001 |
| $S_8$ | $\{2,5,11,13\}$ | $\{1,3,4,5,12,13,15,16,19,23\}$ | $\{2,6,7,8,9,10,11,14,17,18,20,21,22,24\}$ | 010001111110 010011011101 |
| $S_9$ | $\{2,6,9,16\}$ | $\{1,5,9,10,12,15,17,21,22,24\}$ | $\{2,3,4,6,7,8,11,13,14,16,18,19,20,23\}$ | 011101110010 110101110010 |
| $S_{10}$ | $\{2,6,10,15\}$ | $\{1,5,7,8,12,15,17,19,20,24\}$ | $\{2,3,4,6,9,10,11,13,14,16,18,21,22,23\}$ | 011101001110 110101001110 |
| $S_{11}$ | $\{2,8,10,15\}$ | $\{1,5,7,8,12,14,18,19,20,22\}$ | $\{2,3,4,6,9,10,11,13,15,16,17,21,23,24\}$ | 011101001110 101110001011 |
| $S_{12}$ | $\{2,8,11,13\}$ | $\{1,3,4,5,12,14,15,16,18,22\}$ | $\{2,6,7,8,9,10,11,13,17,19,20,21,23,24\}$ | 010001111110 100010111011 |
| $S_{13}$ | $\{3,5,10,14\}$ | $\{2,5,6,8,9,13,17,18,19,23\}$ | $\{1,3,4,7,10,11,12,14,15,16,20,21,22,24\}$ | 101100100111 011100011101 |
| $S_{14}$ | $\{3,5,11,13\}$ | $\{2,3,4,8,9,13,15,16,19,23\}$ | $\{1,5,6,7,10,11,12,14,17,18,20,21,22,24\}$ | 100011100111 010011011101 |
| $S_{15}$ | $\{3,7,9,17\}$ | $\{2,8,9,11,12,16,20,21,23,24\}$ | $\{1,3,4,5,6,7,10,13,14,15,17,18,19,22\}$ | 101111100100 111011100100 |
| $S_{16}$ | $\{3,7,10,14\}$ | $\{2,5,6,8,9,16,17,18,20,21\}$ | $\{1,3,4,7,10,11,12,13,14,15,19,22,23,24\}$ | 101100100111 111000100111 |
| $S_{17}$ | $\{3,8,9,17\}$ | $\{2,8,9,11,12,14,18,22,23,24\}$ | $\{1,3,4,5,6,7,10,13,15,16,17,19,20,21\}$ | 101111100100 101110111000 |
| $S_{18}$ | $\{3,8,11,13\}$ | $\{2,3,4,8,9,14,15,16,18,22\}$ | $\{1,5,6,7,10,11,12,13,17,19,20,21,23,24\}$ | 100011100111 100010111011 |
| $S_{19}$ | $\{4,6,10,15\}$ | $\{4,6,7,8,10,15,17,19,20,24\}$ | $\{1,2,3,5,9,11,12,13,14,16,18,21,22,23\}$ | 111010001011 110101001110 |
| $S_{20}$ | $\{4,6,11,12\}$ | $\{1,2,4,6,10,13,14,15,17,24\}$ | $\{3,5,7,8,9,11,12,16,18,19,20,21,22,23\}$ | 001010111011 000101111110 |
| $S_{21}$ | $\{4,7,9,17\}$ | $\{4,6,10,11,12,16,20,21,23,24\}$ | $\{1,2,3,5,7,8,9,13,14,15,17,18,19,22\}$ | 111010111000 111011100100 |
| $S_{22}$ | $\{4,7,11,12\}$ | $\{1,2,4,6,10,13,14,16,20,21\}$ | $\{3,5,7,8,9,11,12,15,17,18,19,22,23,24\}$ | 001010111011 001011100111 |
| $S_{23}$ | $\{4,8,9,17\}$ | $\{4,6,10,11,12,14,18,22,23,24\}$ | $\{1,2,3,5,7,8,9,13,15,16,17,19,20,21\}$ | 111010111000 101110111000 |
| $S_{24}$ | $\{4,8,10,15\}$ | $\{4,6,7,8,10,14,18,19,20,22\}$ | $\{1,2,3,5,9,11,12,13,15,16,17,21,23,24\}$ | 111010001011 101110001011 |

rule yields exactly the 48 completion vectors in Table IV. It is a recipe for this specific example; the general existence assertion is established independently by Lemma 3.

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TABLE X. Vertex state profiles and weights for all 65 vertices of $H_{65}$, showing for each core vertex $i \in \{1,\dots,17\}$ and each completion pair $(w_{m,1}, w_{m,2}) = (2m+16, 2m+17)$ $(m = 1,\dots,24)$ the number of active branches $W_{24}$, the weight $W_{48} = \sum_{\alpha,\varepsilon} s_{\alpha,\varepsilon}(i)$ across the 48 canonical states, the weight $W_{393,216} = \sum_{s\in\mathcal{S}(H_{65})} s(i)$ across all 393,216 states, the active branch indices $\alpha \in \{1,\dots,24\}$ (equal to $Z(e_m)$ for $(w_{m,1}, w_{m,2})$), and the 24-bit branch profile across $S_1,\dots,S_{24}$ (in which each `1` expands to `11` for core vertices $i$, to `10` for $w_{m,1}$, and to `01` for $w_{m,2}$ across the 48 canonical states).

| Vertex / completion pair | $W_{24}$ | $W_{48}$ | $W_{393,216}$ | Active branches $\alpha \in \{1,\dots,24\}$ | 24-bit branch profile ($1\dots8 \mid 9\dots16 \mid 17\dots24$) |
|---|---|---|---|---|---|
| Core 1 | 6 | 12 | 98,304 | $\{1,2,3,4,5,6\}$ | 11111100 00000000 00000000 |
| Core 2 | 6 | 12 | 98,304 | $\{7,8,9,10,11,12\}$ | 00000011 11110000 00000000 |
| Core 3 | 6 | 12 | 98,304 | $\{13,14,15,16,17,18\}$ | 00000000 00001111 11000000 |
| Core 4 | 6 | 12 | 98,304 | $\{19,20,21,22,23,24\}$ | 00000000 00000000 00111111 |
| Core 5 | 6 | 12 | 98,304 | $\{1,2,7,8,13,14\}$ | 11000011 00001100 00000000 |
| Core 6 | 6 | 12 | 98,304 | $\{3,4,9,10,19,20\}$ | 00110000 11000000 00110000 |
| Core 7 | 6 | 12 | 98,304 | $\{5,6,15,16,21,22\}$ | 00001100 00000011 00001100 |
| Core 8 | 6 | 12 | 98,304 | $\{11,12,17,18,23,24\}$ | 00000000 00110000 11000011 |
| Core 9 | 8 | 16 | 131,072 | $\{1,3,7,9,15,17,21,23\}$ | 10100010 10000010 10001010 |
| Core 10 | 8 | 16 | 131,072 | $\{2,5,10,11,13,16,19,24\}$ | 01001000 01101001 00100001 |
| Core 11 | 8 | 16 | 131,072 | $\{4,6,8,12,14,18,20,22\}$ | 00010101 00010100 01010100 |
| Core 12 | 4 | 8 | 65,536 | $\{4,6,20,22\}$ | 00010100 00000000 00010100 |
| Core 13 | 4 | 8 | 65,536 | $\{8,12,14,18\}$ | 00000001 00010100 01000000 |
| Core 14 | 4 | 8 | 65,536 | $\{2,5,13,16\}$ | 01001000 00001001 00000000 |
| Core 15 | 4 | 8 | 65,536 | $\{10,11,19,24\}$ | 00000000 01100000 00100001 |
| Core 16 | 4 | 8 | 65,536 | $\{1,3,7,9\}$ | 10100010 10000000 00000000 |
| Core 17 | 4 | 8 | 65,536 | $\{15,17,21,23\}$ | 00000000 00000010 10001010 |
| $E_1$ : (18, 19) | 14 | 14 | 114,688 | $\{1,2,3,5,13,14,15,16,17,18,19,21,23,24\}$ | 11101000 00001111 11101011 |
| $E_2$ : (20, 21) | 14 | 14 | 114,688 | $\{1,2,3,5,7,8,9,10,11,12,19,21,23,24\}$ | 11101011 11110000 00101011 |
| $E_3$ : (22, 23) | 14 | 14 | 114,688 | $\{7,9,10,11,13,15,16,17,19,20,21,22,23,24\}$ | 00000010 11101011 10111111 |
| $E_4$ : (24, 25) | 14 | 14 | 114,688 | $\{1,2,3,4,5,6,7,9,10,11,13,15,16,17\}$ | 11111110 11101011 10000000 |
| $E_5$ : (26, 27) | 14 | 14 | 114,688 | $\{1,3,4,6,14,15,17,18,19,20,21,22,23,24\}$ | 10110100 00000110 11111111 |
| $E_6$ : (28, 29) | 14 | 14 | 114,688 | $\{1,3,4,6,7,8,9,10,11,12,14,15,17,18\}$ | 10110111 11110110 11000000 |
| $E_7$ : (30, 31) | 14 | 14 | 114,688 | $\{7,8,9,12,13,14,15,16,17,18,20,21,22,23\}$ | 00000011 10011111 11011110 |
| $E_8$ : (32, 33) | 14 | 14 | 114,688 | $\{1,2,3,4,5,6,7,8,9,12,20,21,22,23\}$ | 11111111 10010000 00011110 |
| $E_9$ : (34, 35) | 14 | 14 | 114,688 | $\{2,4,5,6,8,10,11,12,19,20,21,22,23,24\}$ | 01011101 01110000 00111111 |
| $E_{10}$ : (36, 37) | 14 | 14 | 114,688 | $\{2,4,5,6,8,10,11,12,13,14,15,16,17,18\}$ | 01011101 01111111 11000000 |
| $E_{11}$ : (38, 39) | 14 | 14 | 114,688 | $\{7,8,9,10,11,12,13,14,16,18,19,20,22,24\}$ | 00000011 11111101 01110101 |
| $E_{12}$ : (40, 41) | 14 | 14 | 114,688 | $\{1,2,3,4,5,6,13,14,16,18,19,20,22,24\}$ | 11111100 00001101 01110101 |
| $E_{13}$ : (42, 43) | 14 | 14 | 114,688 | $\{3,5,9,10,11,12,15,16,17,18,19,21,23,24\}$ | 00101000 11110011 11101011 |
| $E_{14}$ : (44, 45) | 14 | 14 | 114,688 | $\{1,2,3,5,7,8,9,10,13,14,15,16,19,21\}$ | 11101011 11001111 00101000 |
| $E_{15}$ : (46, 47) | 14 | 14 | 114,688 | $\{1,2,5,6,7,11,13,15,16,17,21,22,23,24\}$ | 11001110 00101011 10001111 |
| $E_{16}$ : (48, 49) | 14 | 14 | 114,688 | $\{1,2,3,4,7,9,10,11,13,17,19,20,23,24\}$ | 11110010 11101000 10110011 |
| $E_{17}$ : (50, 51) | 14 | 14 | 114,688 | $\{1,6,7,8,11,12,14,15,17,18,21,22,23,24\}$ | 10000111 00110110 11001111 |
| $E_{18}$ : (52, 53) | 14 | 14 | 114,688 | $\{1,3,4,6,7,8,9,10,14,15,19,20,21,22\}$ | 10110111 11000110 00111100 |
| $E_{19}$ : (54, 55) | 14 | 14 | 114,688 | $\{3,4,5,6,9,12,15,16,17,18,20,21,22,23\}$ | 00111100 10010011 11011110 |
| $E_{20}$ : (56, 57) | 14 | 14 | 114,688 | $\{1,2,3,4,7,8,9,12,13,14,17,18,20,23\}$ | 11110011 10011100 11010010 |
| $E_{21}$ : (58, 59) | 14 | 14 | 114,688 | $\{2,4,8,10,11,12,13,14,17,18,19,20,23,24\}$ | 01010001 01111100 11110011 |
| $E_{22}$ : (60, 61) | 14 | 14 | 114,688 | $\{2,4,5,6,8,10,13,14,15,16,19,20,21,22\}$ | 01011101 01001111 00111100 |
| $E_{23}$ : (62, 63) | 14 | 14 | 114,688 | $\{3,4,5,6,9,10,11,12,16,18,19,20,22,24\}$ | 00111100 11110001 01110101 |
| $E_{24}$ : (64, 65) | 14 | 14 | 114,688 | $\{1,2,5,6,7,8,11,12,13,14,16,18,22,24\}$ | 11001111 00111101 01000101 |